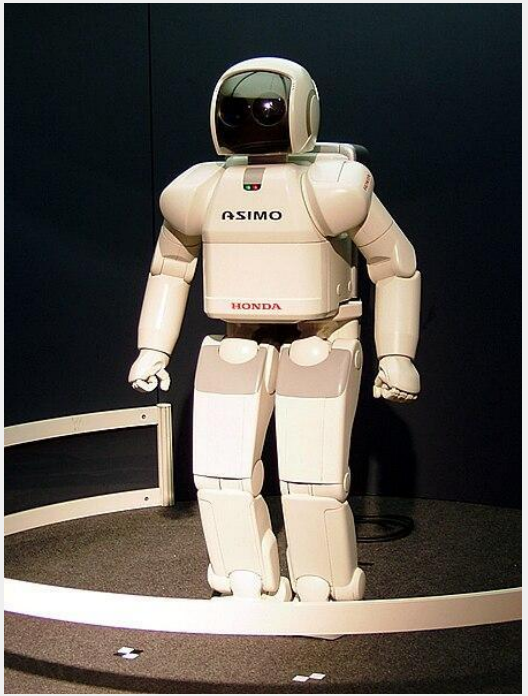
History of AI

Global Perspectives in History of AI



unsojo@cornell.edu

Lecture 4 Februrary 14, 2025

CNBC MARKETS BUSINESS INVESTING TECH POLITICS VIDEO INVESTING CLUB JOIN PRO JOIN LIVESTREAM

TECH

Musk will withdraw OpenAI bid if ChatGPT maker stays nonprofit, lawyers say

PUBLISHED THU, FEB 13 2025•7:00 AM EST | UPDATED 5 HOURS AGO

Natasha Turak
@NATASHATURAK

SHARE

KEY POINTS

- Musk and a consortium of investors previously launched a bid to acquire OpenAI's nonprofit arm for $97.4 billion.
- OpenAI CEO Sam Altman has rebuffed the offer, telling CNBC that the move is just an effort by Musk to "slow down a competitor."
- OpenAI was initially founded as a nonprofit in 2015, and later converted to a "capped profit" model in 2019.



Prefer to Listen
NOW
UP NEXT

Business
Premier

FORBES › BUSINESS

BREAKING

Sam Altman-Elon Musk Feud Timeline: Altman Says Musk Isn't A 'Happy Person' After 'Crazy' OpenAI Offer

Derek Saul Forbes Staff
Derek Saul has covered markets for the Forbes news team since 2021.

Follow

0

Feb 11, 2025, 08:59am EST

TOPLINE The dispute between arguably the two biggest names in generative artificial intelligence—billionaires Sam Altman and Elon Musk—received a new wrinkle Monday as Musk made a nearly $100 billion offer to buy the non-profit entity controlling Altman's OpenAI, though the feud quickly devolved again into name-calling as the billionaire beef bleeds into boardrooms, courts and the Oval Office.



Elon Musk, left, and Sam Altman appear together in 2015. GETTY IMAGES FOR VANITY FAIR

TIMELINE

- February 10, 2025

TECH· A.I.

Marc Andreessen warns Chinese ChatGPT rival DeepSeek is 'AI's Sputnik moment'

BY CHRISTIAAN HETZNER
January 27, 2025 at 9:00 AM EST



Venture capitalist Marc Andreessen sees the release of R1 by a Chinese AI lab as comparable to the Soviet Union's launch of the first satellite into orbit.
DAVID PAUL MORRIS—BLOOMBERG/GETTY IMAGES

AI History

PYMNTS

SEARCH

PYMNTS TV | Today | B2B | Retail | Fintech | Digital Transformation | Crypto | AI | PYMNTS® Intelligence

AI Summit: EU, France to Invest $320 Billion in AI to Rival US, China

TREND

Energy, Increas

Coupa E Spend M

BY **PYMNTS** | FEBRUARY 11, 2025



This week's AI Action Summit in Paris — the artificial intelligence (AI) comm
— saw the Europeans act decidedly un-European.

ALJAZEERA

News | Middle East | Explained | Opinion | Sport | Video | M

Trending | Russia-Ukraine war | Gaza | Donald Trump | Occupied Wes

Vance rails against AI regulation in Paris as US faces off with EU, China

US vice president warns global leaders against striking AI deals with 'authoritarian regimes' in a veiled jab at China.



e delivers a speech during the plenary session of the Artificial Intelligence (AI) Action Summit at enoit Tessier/Reuters]

myNEWS | Latest | China | Economy | HK | Asia | Business | Tech | Lifestyle | People & Culture | World | Opinion | Video | Sport | PostMag | Style | All

Artificial intelligence | *Tech / Big Tech*

DeepSeek founder Liang Wenfeng skips Paris AI summit as China champions its global role

China's academics tout the country's contribution to global AI development, while its hottest start-up avoids the limelight

Reading Time: **2 minutes**

Why you can trust SCMP

Listen



1. No.1 **AI** Stock to $10
2. Elon Musk's **No.1** Buy Now
3. 10 **Stocks** to **Hol**

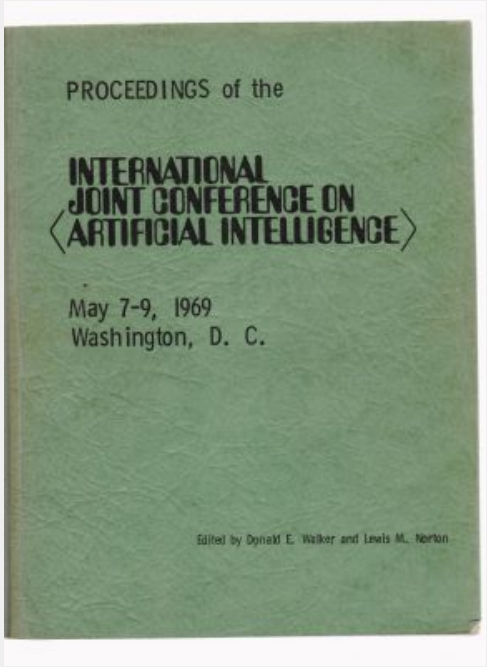
Ad | Select Searches

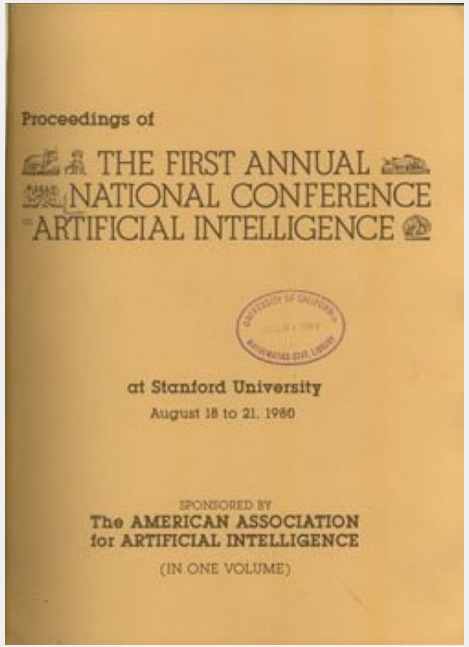
Last Week!

- ARPA/DARPA in Cold War U.S.
- Rule-based AI + LISP
- Interactive Computing
- Ethics of Autonomous Weapons

AI Conferences & Expos

International Joint Conference on Artificial Intelligence (IJCAI) 1969







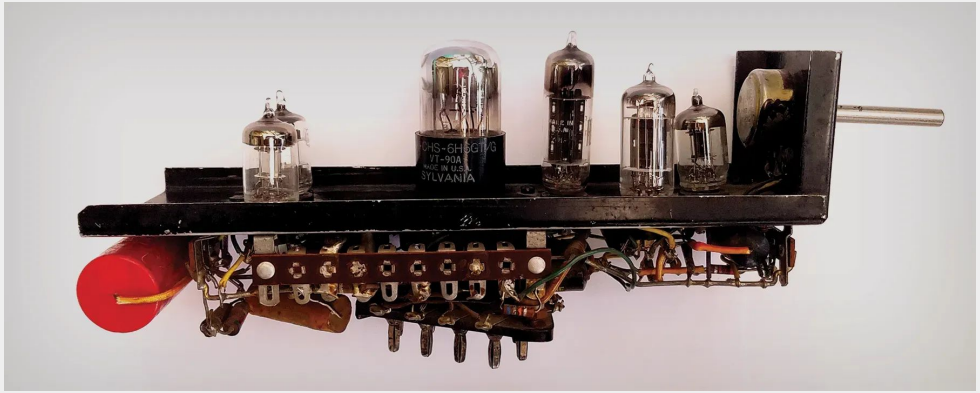
Consumer Electronics Show 1967

AAAI Conference (1980)

European Conference on AI (ECAI) [previously AISB] 1974

Marvin Minsky's SNARC 1951

Pitts-McCulloch neural nets + B.F. Skinner reinforcement learning



1 neuron!

- made out of b-24 bomber aircraft pieces
- 40 neurons
- 6 vacuum tubes each
- size of small room
- was able to solve Shannon's maze
- → sufficient memory for reasoning would require millions of neurons

October 4, 1957

"All the News That's Fit to Print"

The New York Times.

LATE CITY EDITION

VOL. CVII.. No. 36,454. NEW YORK, SATURDAY, OCTOBER 5, 1957. FIVE CENTS

SOVIET FIRES EARTH SATELLITE INTO SPACE; IT IS CIRCLING THE GLOBE AT 18,000 M. P. H.; SPHERE TRACKED IN 4 CROSSINGS OVER U. S.

HOFFA IS ELECTED TEAMSTERS' HEAD; WARNS OF BATTLE

Defeats Two Foes 3 to 1 —Says Union Will Fight 'With Every Ounce'



IN TOKEN OF VICTORY: Dave Beck, retiring head of the Teamsters Union, raises hand of James R. Hoffa upon his election as union's president. At right is Mrs. Hoffa.

COURSE RECORDED

Navy Picks Up Radio Signals—4 Report Sighting Device



Device Is 8 Times Heavier Than One Planned by U.S.

560 MILES HIGE

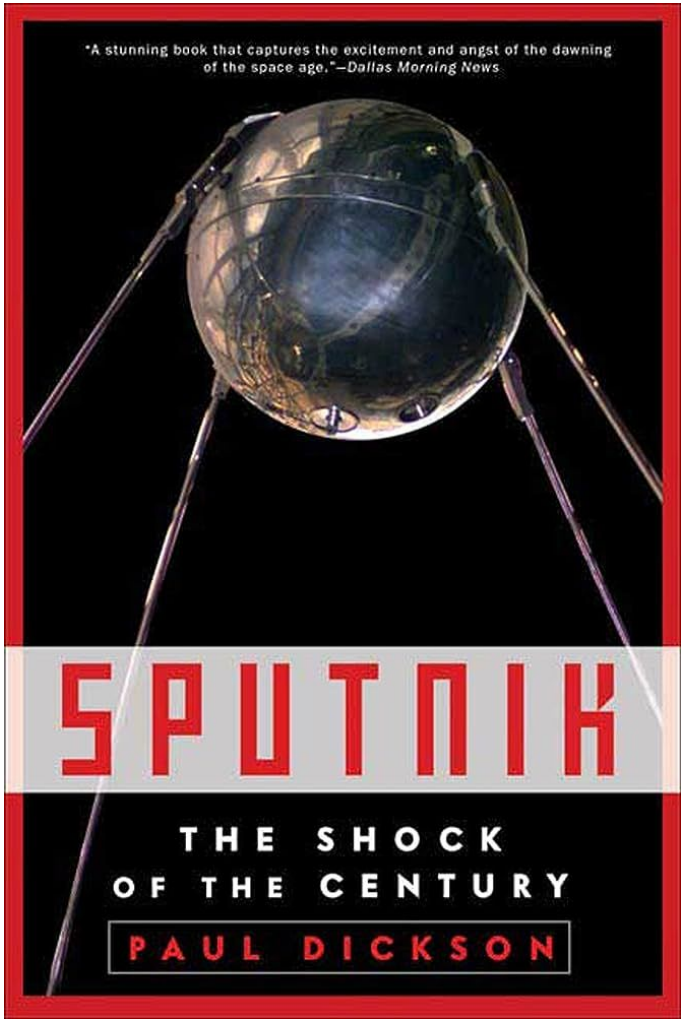
Visible With Simple Binoculars, Moscow Statement Says

FAUBUS COMPARES HIS STAND TO LEE'S

Flu Widens in City; 10% Rate Predicted; 200,000 Pupils Out

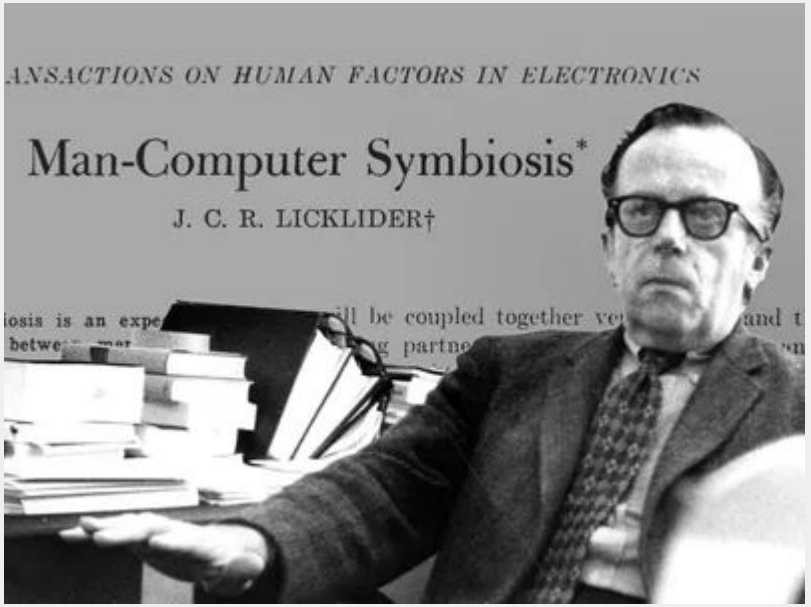
ARGENTINA TAKES EMERGENCY STEPS

SATELLITE SIGNAL

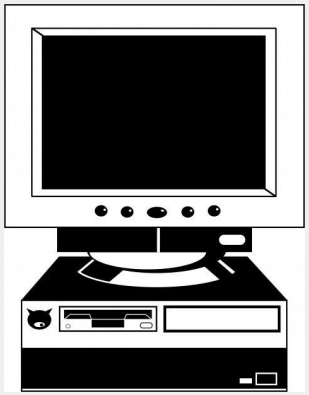


ARPA's IPTO 1961
Information Processing Techniques Office

- JFK's directive to step up "command and control" research
- Headed by "Lick" J.C.R. Licklider, psychologist who studied computer interaction & psychoacoustics in the 1950s
- "the electronic members of the partnership"
- **"Command and control" included research in ballistic missiles, radar tracking, high-energy lasers, etc.**



“Lick’s” Man-Computer Symbiosis 1960





speed and accuracy

flexibility and intuition

“The hope is that, in not too many years, human brains and computing machines will be coupled together very tightly, and that the resulting partnership will think as no human brain has ever thought and process data in a new way not approached by the information-handling machines we know today.”

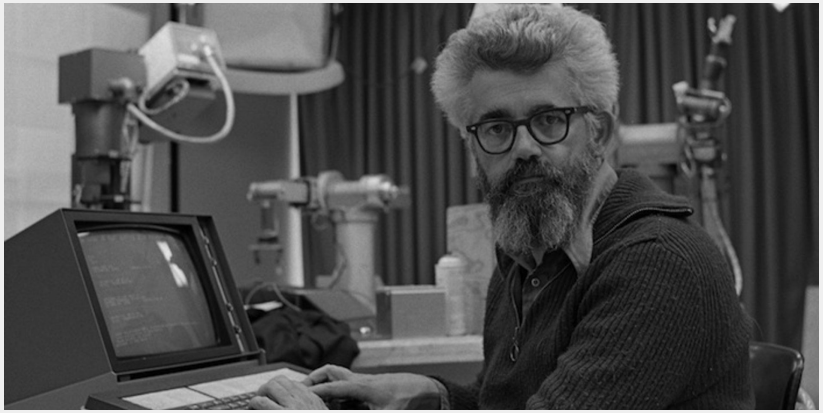
JCR Licklider, Man-Computer Symbiosis, 1960

JCR Licklider



ARPA/DARPA funds U.S. AI institutions

- MIT: Marvin Minsky and students
- Stanford: John McCarthy started the Stanford AI Lab with unconditional ARPA money 1964
- ARPA supported AI at Rand, Carnegie-Mellon, SRI, SDC (Systems Development Corp) ,BBN (Bolt, Beranak, Newman), and seven other “centers of excellence”
- ARPA became the primary patron for the first twenty years of AI research

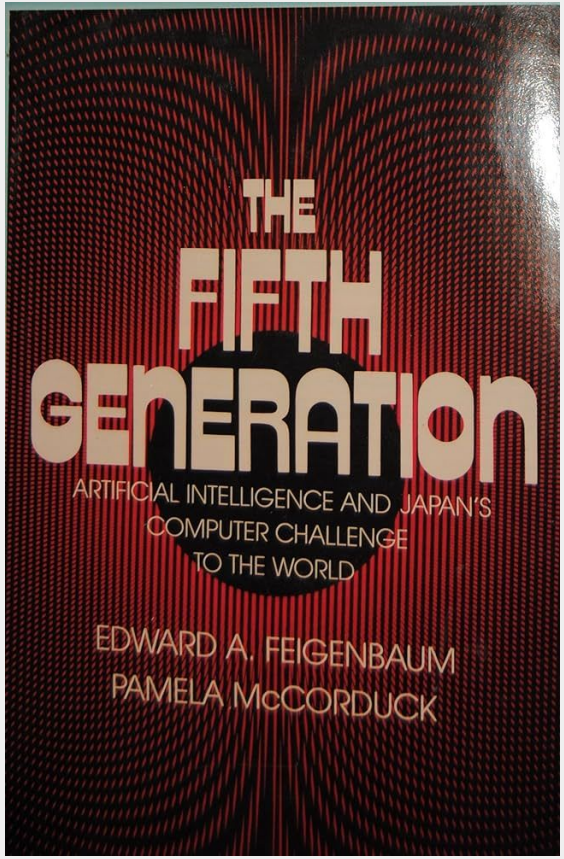


Strategic Computing Initiative

- 1 decade long (1983-93)
- 1 billion dollars
- Inspired/threatened by Japan's (MITI's) Fifth Generation Computer Program (1982-, $800million dollar budget)
- By 1985, DARPA had spent $100 million for 92 projects underway at 60 institutions (half in industry, half in universities and gov)
- SCI planners made a detailed ten-year timeline to coordinate all of these subprojects to come together to create 3 intelligent projects

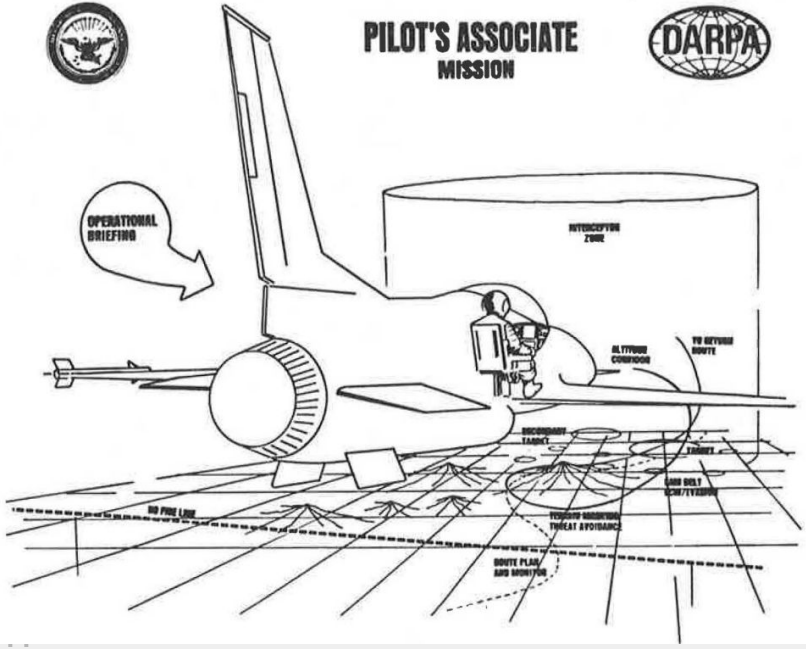
SC

STRATEGIC COMPUTING





Pilot's Associate Mission.

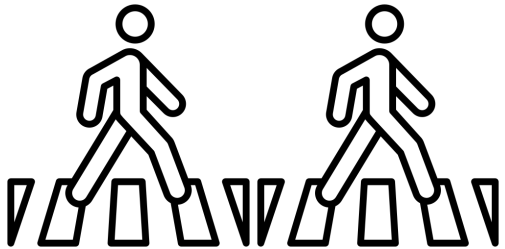


Trolley Problem for Machines

Autonomous car has to decide colliding with single pedestrian vs. multiple pedestrians or to save the car's passenger.







Language AI

- Why language AI?
- Is communication our basis/standard of intelligence?
- How does commercialization of technology change its development?

AI or not?



AI or not?



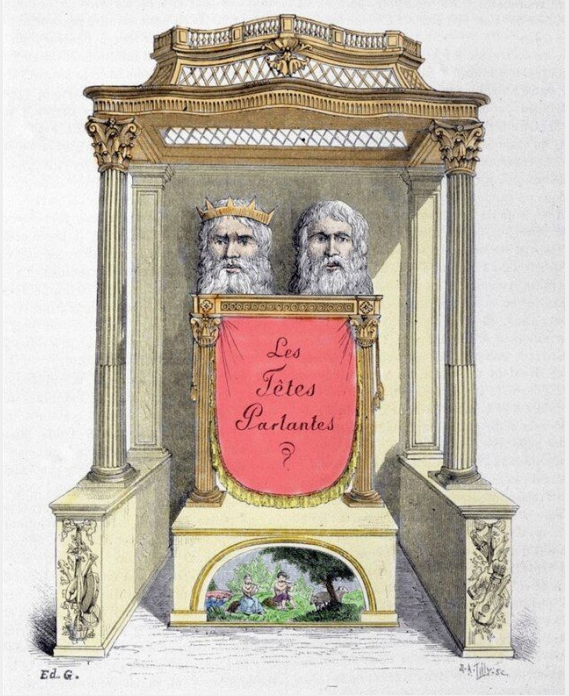
AI or not?



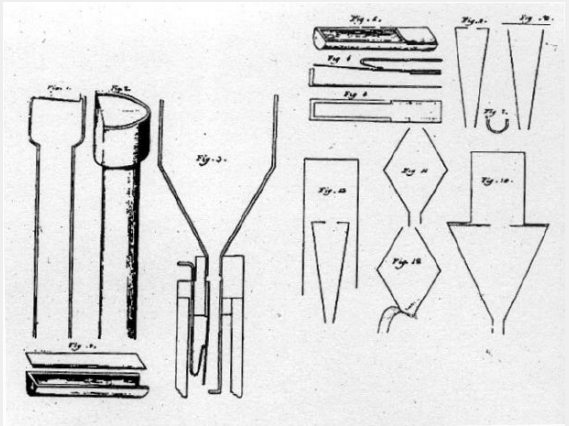
AI or not?



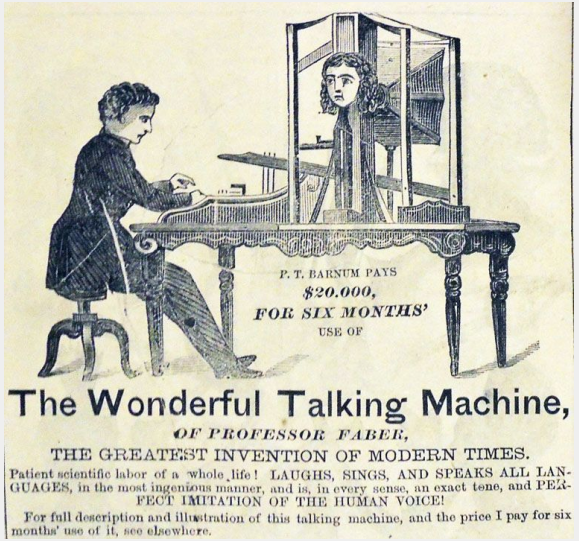
Why language AI? Early Speech Simulation (not synthesis)



abbe Mical's Talking Heads. Paris Academy of Sciences. 1778.

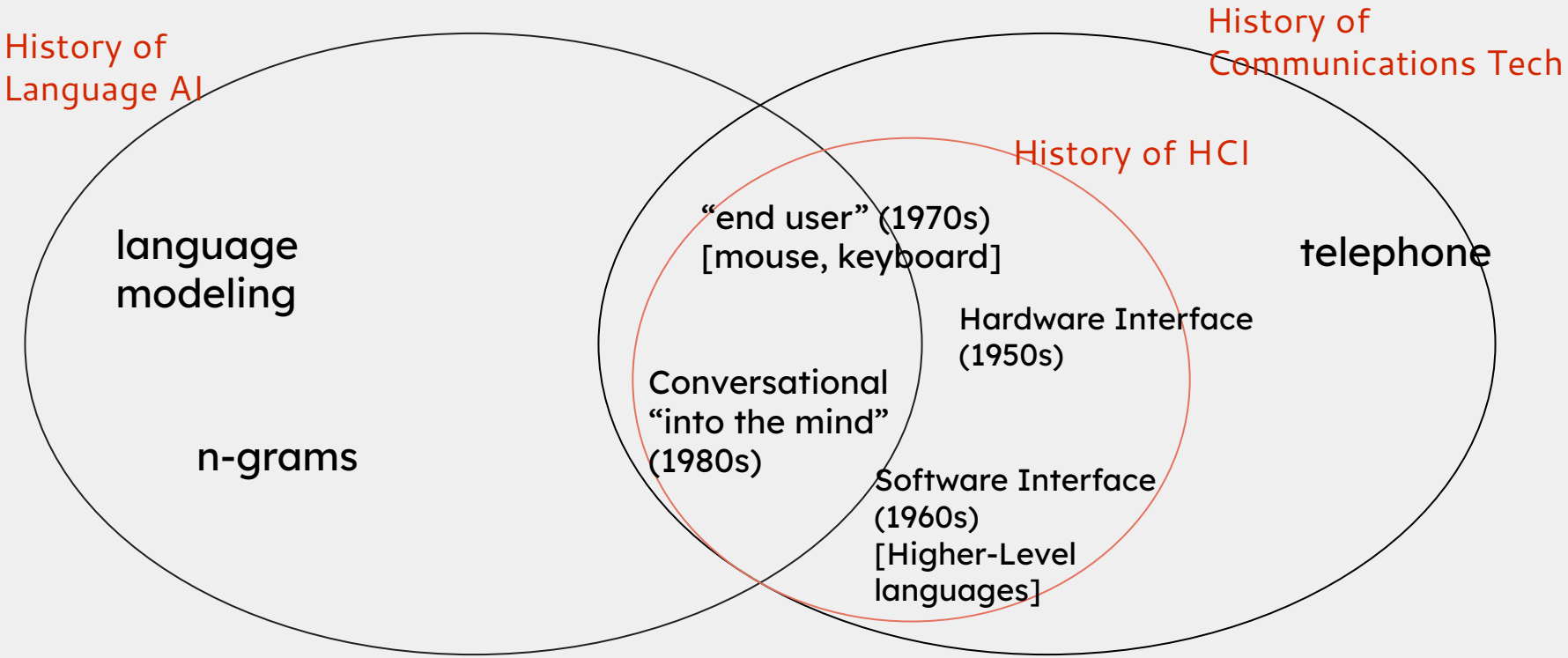


C. G. Kratzenstein. *"vox humana"* Saint Petersburg Academy of Sciences. 1784.



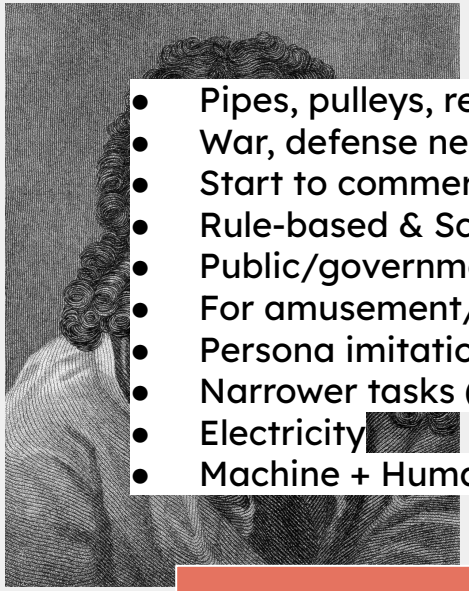
Joseph Faber. Talking Machine. 1840s.

Language AI vs. Communication Tech



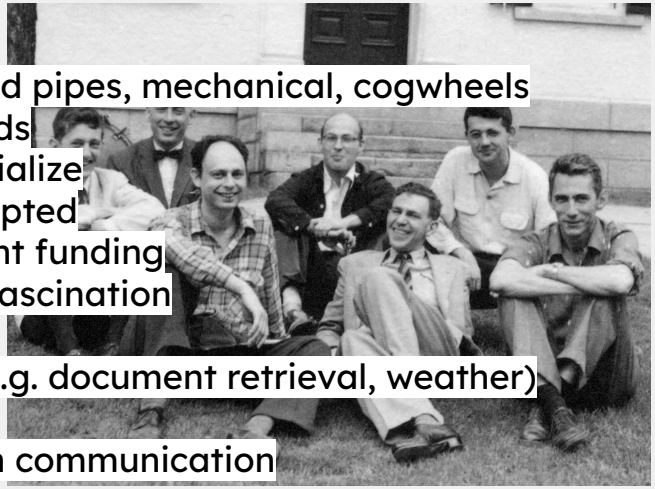
Language AI

Early Modern

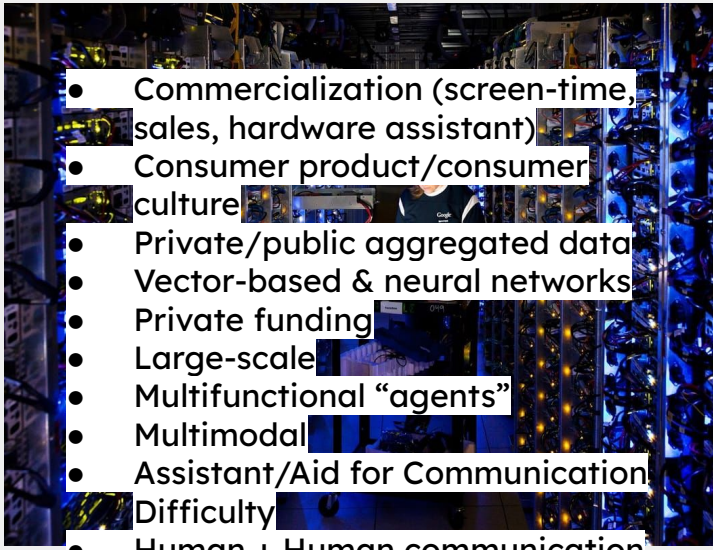


- Pipes, pulleys, reed pipes, mechanical, cogwheels
- War, defense needs
- Start to commercialize
- Rule-based & Scripted
- Public/government funding
- For amusement/fascination
- Persona imitation
- Narrower tasks (e.g. document retrieval, weather)
- Electricity
- Machine + Human communication

Modern



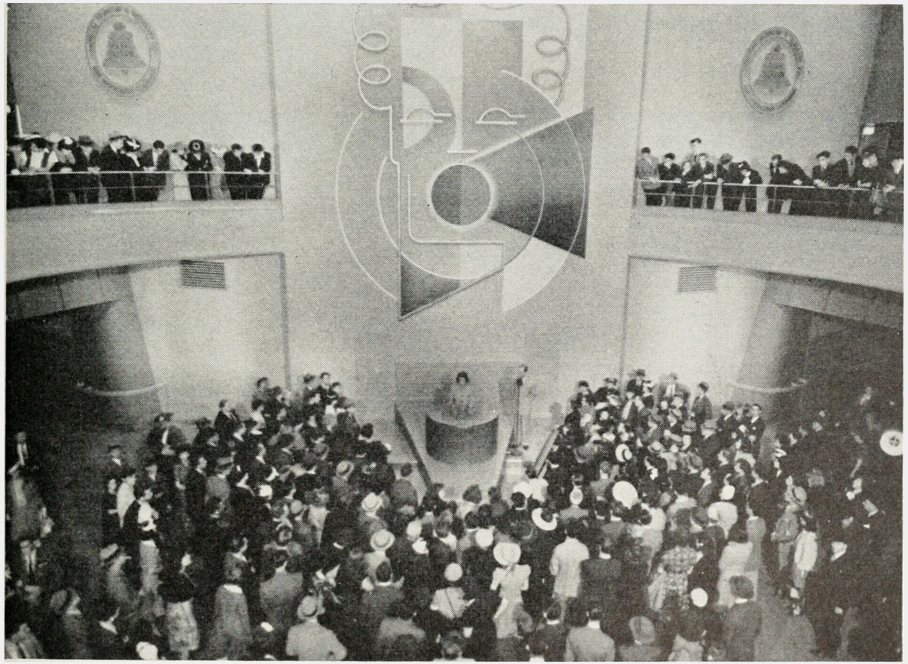
Post-Internet AI



- Commercialization (screen-time, sales, hardware assistant)
- Consumer product/consumer culture
- Private/public aggregated data
- Vector-based & neural networks
- Private funding
- Large-scale
- Multifunctional “agents”
- Multimodal
- Assistant/Aid for Communication Difficulty
- Human + Human communication

Speech before NLP: World Fair 1939

- Speech simulation → Synthesis
- Generating waveforms of audio to simulate human voice
- Bell Labs (AT&T) first experiments in electronic voice synthesis
- Voder (Voice Demonstrator)
- Machine Age → Information Age
- Telephone ownership about 40% 1930s





Speech Synthesis

- AT&T Bell Labs developed a digits recognizing speech system called AUDREY (1952) through simple acoustic pattern matching. AUDREY could understand numbers.
- Transistors → Big Increase in voice technology
- First voice businesses appeared
- 1962 IBM presented the program SHOEBOX, that could understand 16 words
- 1962 Am-Quote that could read stock information



AUDREY 1952

Machine Translation 1947

- Post WWII, machine translation becomes a defense agenda
- Mechanical/Manual translation too time-consuming and inaccurate

One thing I wanted to ask you about is this. A most serious problem, for UNESCO and for the constructive and peaceful future of the planet, is the problem of *translation*, as it unavoidably affects the communication between peoples. Huxley has recently told me that they are appalled by the magnitude and the importance of the translation job.

Recognizing fully, even though necessarily vaguely, the semantic difficulties because of multiple meanings, etc., I have wondered if it were unthinkable to design a computer which would translate. Even if it would translate only scientific material (where the semantic difficulties are very notably less), and even if it did produce an inelegant (but intelligible) result, it would seem to me worth while.

Also knowing nothing official about, but having guessed and inferred considerable about, powerful new mechanized methods in cryptography – methods which I believe succeed even when one does not know what language has been coded – one naturally wonders if the problem of translation could conceivably be treated as a problem in cryptography. When I look at an article in Russian, I say "This is really written in English, but it has been coded in some strange symbols. I will now proceed to decode."

Have you ever thought about this? As a linguist and expert on computers, do you think it is worth thinking about?

March 1947, Warren Weaver, director of the Rockefeller Foundation writing to Cryptographer friend Norbert Wiener (the cybernetician)

“Mechanical” Translation’s Problems

- With “mechanical” translation, a word-for-word dictionary search via a computer, two problems arose:

1. memory:

One way of dealing with this was to limit the size of the vocabulary to the most common words.

2. syntax and word disambiguation:

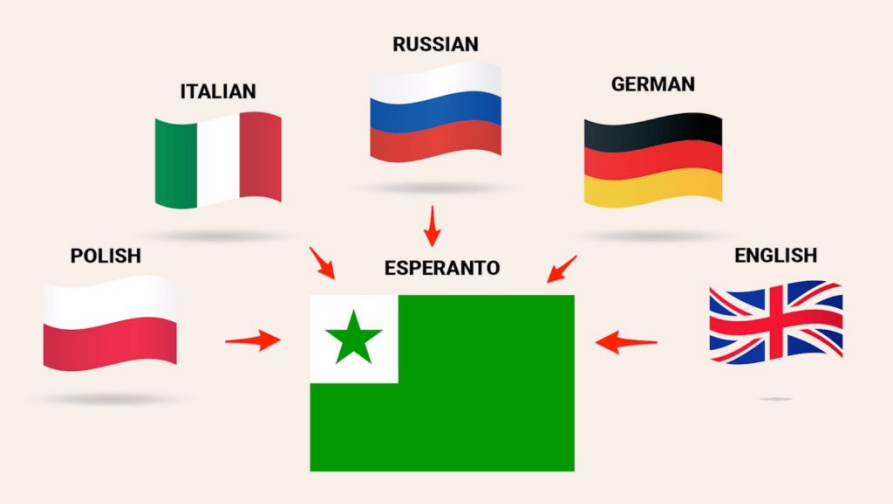
Some proposed an “international language” or semantic blocks that could encode the essence of the input and translate into the output language– “common” language such as Esperanto to capture the shared human meaning of input and output languages.

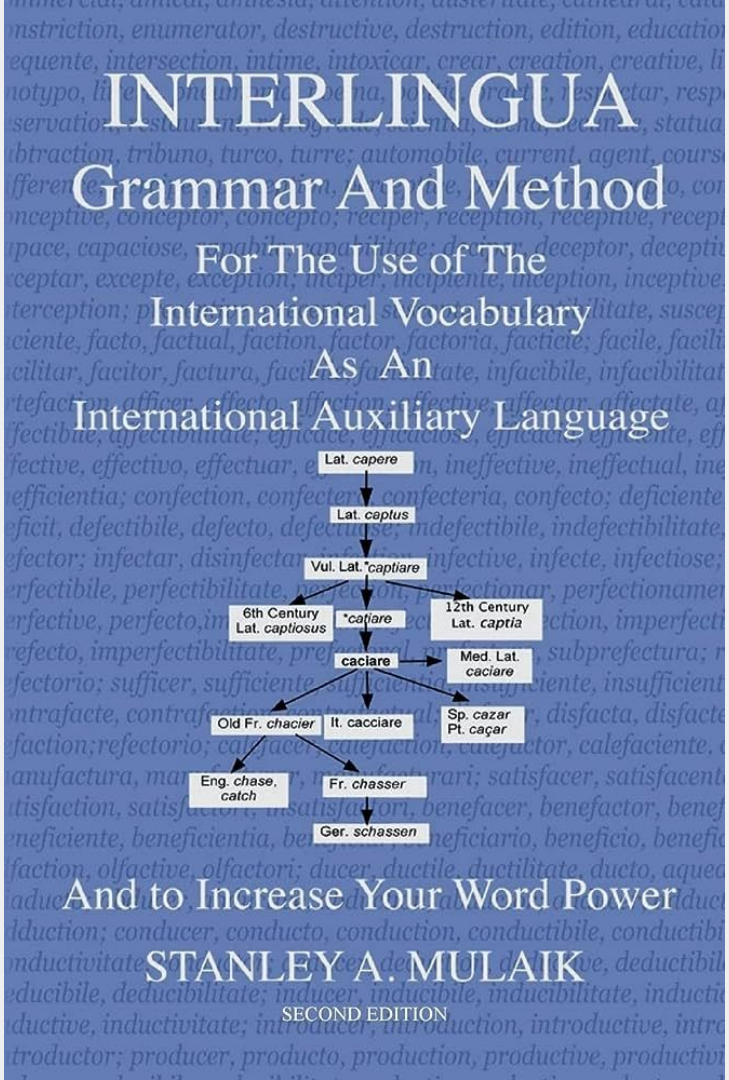
1948 Segmentation

(German) Wenn in ein*em gröss*eren Gebiet zwei Form*en neben*einander leb*en, ohne sich zu vermisch*en, so gehör*en sie verschieden*en Form*en*-kreis*en an.
(English) if in a/one *d* large (more) area two form *m* beside one another live *z* without self to/too mix *z*, so belong/hear *p z z* different *m* form *m* circle *m* at.

A Global Language

- Politically, debate ensued about the "scientific" or "international" language. What common language would scientists speak to share knowledge globally? English? Esperanto? Interlingua?





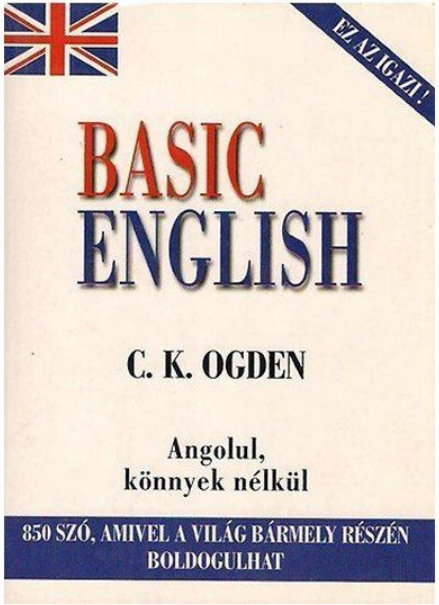
BASIC ENGLISH

OPERATIONS etc.	THINGS						QUALITIES	
100	400 General				200 Pictrured		100 General	50 Opposite
COME	ACCOUNT	EDUCATION	METAL	SENSE	ANGLE	KNEE	ABLE	AWAKE
GET	ACT	EFFECT	MIDDLE	SERVANT	ANT	KNIFE	ACID	BAD
GIVE	ADDITION	END	MILK	SEX	APPLE	KNOT	ANGRY	BENT
GO	ADJUSTMENT	ERROR	MIND	SHADE	ARCH	LEAF	AUTOMATIC	BITTER
KEEP	ADVERTISEMENT	EVENT	MINE	SHAKE	ARM	LEG	BEAUTIFUL	BLUE
LET	AGREEMENT	EXAMPLE	MINUTE	SHAME	ARMY	LIBRARY	BLACK	CERTAIN
MAKE	AIR	EXCHANGE	MIST	SHOCK	BABY	LINE	BOILING	COLD
PUT	AMOUNT	EXISTENCE	MONEY	SIDE	BAG	LIP	BRIGHT	COMPLETE
SEEM	AMUSEMENT	EXPANSION	MONTH	SIGN	BALL	LOCK	BROKEN	CRUEL
TAKE	ANIMAL	EXPERIENCE	MORNING	SILK	BAND	MAP	BROWN	DARK
BE	ANSWER	EXPERT	MOTHER	SILVER	BASIN	MATCH	CHEAP	DEAD
DO	APPARATUS	FACT	MOTION	SISTER	BASKET	MONKEY	CHEMICAL	DEAR
HAVE	APPROVAL	FALL	MOUNTAIN	SIZE	BATH	MOON	CHIEF	DELICATE
SAY	ARGUMENT	FAMILY	MOVE	SKY	BED	MOUTH	CLEAN	DIFFERENT
SEE	ART	FATHER	MUSIC	SLEEP	BEE	MUSCLE	CLEAR	DIRTY
SEND	ATTACK	FEAR	NAME	SLIP	BELL	NAIL	COMMON	DRY
MAY	ATTEMPT	FEELING	NATION	SLOPE	BERRY	NECK	COMPLEX	FALSE
WILL	ATTENTION	FICTION	NEED	SMASH	BIRD	NEEDLE	CONSCIOUS	FEEBLE
ABOUT	ATTRACTION	FIELD	NEWS	SMELL	BLADE	NERVE	CUT	FEMALE
ACROSS	AUTHORITY	FIGHT	NIGHT	SMILE	BOARD	NET	DEEP	FOOLISH
AFTER	BACK	FIRE	NOISE	SMOKE	BOAT	NOSE	DEPENDENT	FUTURE
AGAINST	BALANCE	FLAME	NOTE	SNEEZE	BONE	NUT	EARLY	GREEN
AMONG	BASE	FLIGHT	NUMBER	SNOW	BOOK	OFFICE	ELASTIC	ILL
AT	BEHAVIOUR	FLOWER	OBSERVATION	SOAP	BOOT	ORANGE	ELECTRIC	LAST
BEFORE	BELIEF	FOLD	OFFER	SOCIETY	BOTTLE	OVEN	EQUAL	LATE
BETWEEN	BIRTH	FOOD	OIL	SON	BOX	PARCEL	FAT	LEFT
BY	BIT	FORCE	OPERATION	SONG	BOY	PEN	FERTILE	LOOSE
DOWN	BITE	FORM	OPINION	SORT	BRAIN	PENCIL	FIRST	LOUD
FROM	BLOOD	FRIEND	ORDER	SOUND	BRAKE	PICTURE	FIXED	LOW
IN	BLOW	FRONT	ORGANIZATION	SOUP	BRANCH	PIG	FLAT	MIXED
OFF	BODY	FRUIT	ORNAMENT	SPACE	BRICK	PIN	FREE	NARROW
ON	BRASS	GLASS	OWNER	STAGE	BRIDGE	PIPE	FREQUENT	OLD
OVER	BREAD	GOLD	PAGE	START	BRUSH	PLANE	FULL	OPPOSITE
THROUGH	BREATH	GOVERNMENT	PAIN	STATEMENT	BUCKET	PLATE	GENERAL	PUBLIC
TO	BROTHER	GRAIN	PAINT	STEAM	BULB	PLOUGH	GOOD	ROUGH
UNDER	BUILDING	GRASS	PAPER	STEEL	BUTTON	POCKET	GREAT	SAD
UP	BURN	GRIP	PART	STEP	CAKE	POT	GREY	SAFE
WITH	BURST	GROUP	PASTE	STITCH	CAMERA	POTATO	HANGING	SECRET
AS	BUSINESS	GROWTH	PAYMENT	STONE	CARD	PRISON	HAPPY	SHORT
FOR	BUTTER	GUIDE	PEACE	STOP	CART	PUMP	HARD	SHUT
OF	CANVAS	HARBOUR	PERSON	STORY	CARRIAGE	RAIL	HEALTHY	SIMPLE
TILL	CARE	HARMONY	PLACE	STRETCH	CAT	RAT	HIGH	SLOW
THAN	CAUSE	HATE	PLANT	STRUCTURE	CHAIN	RECEIPT	HOLLOW	SMALL
A	CHALK	HEARING	PLAY	SUBSTANCE	CHEESE	RING	IMPORTANT	SOFT
THE	CHANCE	HEAT	PLEASURE	SUGAR	CHEST	ROD	KIND	SOLID
ALL	CHANGE	HELP	POINT	SUGGESTION	CHIN	ROOF	LIKE	SPECIAL
ANY	CLOTH	HISTORY	POISON	SUMMER	CHURCH	ROOT	LIVING	STRANGE
EVERY	COAL	HOLE	POLISH	SUPPORT	CIRCLE	SAIL	LONG	THIN
LITTLE	COLOUR	HOPE	PORTER	SURPRISE	CLOCK	SCHOOL	MALE	WHITE
MUCH	COMFORT	HOUR	POSITION	SWIM	CLOUD	SCISSORS	MARRIED	WRONG
NO	COMMITTEE	HUMOUR	POWDER	SYSTEM	COAT	SCREW	MATERIAL	
OTHER	COMPANY	ICE	POWER	TALK	COLLAR	SEED	MEDICAL	
SOME	COMPARISON	IDEA	PRICE	TASTE	COMB	SHEEP	MILITARY	
SUCH	COMPETITION	IMPULSE	PRINT	TAX	CORD	SHELF	NATURAL	
THAT	CONDITION	INCREASE	PROCESS	TEACHING	COW	SHIP	NECESSARY	
THIS	CONNECTION	INDUSTRY	PRODUCE	TENDENCY	CUP	SHIRT	NEW	
I	CONTROL	INK	PROFIT	TEST	CURTAIN	SHOE	NORMAL	
HE	COOK	INSECT	PROPERTY	THEORY	CUSHION	SKIN	OPEN	
YOU	COPPER	INSTRUMENT	PROSE	THING	DOG	SKIRT	PARALLEL	
WHO	COPY	INSURANCE	PROTEST	THOUGHT	DOOR	SNAKE	PAST	
AND	CORK	INTEREST	PULL	THUNDER	DRAIN	SOCK	PHYSICAL	
BECAUSE	COTTON	INVENTION	PUNISHMENT	TIME	DRAWER	SPADE	POLITICAL	
BUT	COUGH	IRON	PURPOSE	TIN	DRESS	SPONGE	POOR	
OR	COUNTRY	JELLY	PUSH	TOP	DROP	SPOON	POSSIBLE	
IF	COVER	JOIN	QUALITY	TOUCH	EAR	SPRING	PRESENT	
THOUGH	CRACK	JOURNEY	QUESTION	TRADE	EGG	SQUARE	PRIVATE	
WHILE	CREDIT	JUDGE	RAIN	TRANSPORT	ENGINE	STAMP	PROBABLE	
HOW	CRIME	JUMP	RANGE	TRICK	EYE	STAR	QUICK	
WHEN	CRUSH	KICK	RATE	TROUBLE	FACE	STATION	QUIET	
WHERE	CRY	KISS	RAY	TURN	FARM	STEM	READY	

Ogden's English 1930

- Ogden (1930) proposed that most of our thoughts could be encompassed by just 850 words in the English vocabulary.





Georgetown IBM MT Experiments Jan 7th 1954

- IBM/Georgetown researchers put on a public demo
- IBM 701 translated Russian to English
- 250 Russian words, 6 rules of operational syntax
- 60 sentences
- 6-7 seconds per sentences
- CIA, NSA, Office of Naval Research

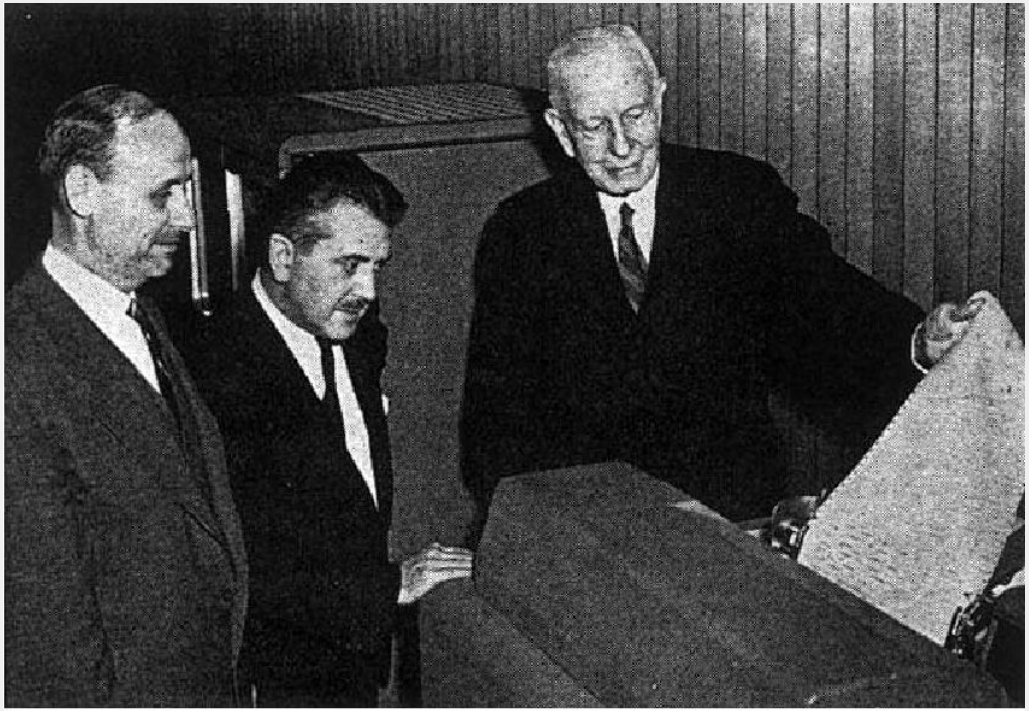


Fig. 2: Hurd, Dostert and Watson at the demonstration

Ruský originál (a angl. transkripce)	Anglický překlad
Качество угля определяется калорийностью Kachyestvo uglya opryedyelyayetsya kaloryiynostjyu	The quality of coal is determined by calory content
Крахмал вырабатывается механическим путем из картофеля Kraxmal virabativayetsya myexanyichyeskyim putyem yiz kartofyelya	Starch is produced by mechanical methods from potatoes
Железо добывается из руды химическим процессом Zhyelyezo dobivayetsya yiz rudi xyimyicheskyim protsyessom	Iron is obtained from ore by chemical process
Мы передаем мысли посредством речи Mi pyeryedayem mislyi posryedstvom ryechi	We transmit thoughts by means of speech
Величина угла определяется отношением длины дуги к радиусу Vyelyichyina ugla opryedyelyayetsya otnoshyenyiyem dlyini dugi k radyiusu	Magnitude of angle is determined by the relation of length of arc to radius
Международное понимание является важным фактором в решении политических вопросов Myezhdunarodnoye ponyimanyiye yavlyayetsya vazhnim faktorom v ryeshyenyiyi polyityichyeskyix voprosov	International understanding constitutes an important factor in decision of political questions
Владимир является на работу поздно утром Vladyimir yavlyayetsya na rabotu pozdno utrom	Vladimir appears for work late in the morning

IBM 701 Electronic Data Processing Machine

- $400,000 machine consisting of eleven separate units and was the size of a tennis court.
- There were only 70 computers in the U.S. in 1954
- Input: IBM 80 column cards



IBM 701. 1954

Russian Is Turned Into English By a Fast Electronic Translator

By ROBERT K. PLUMB

A public demonstration of what is believed to be the first successful use of a machine to translate meaningful texts from one language to another took place here yesterday afternoon.

This may be the culmination of centuries of search by scholars for "a mechanical translator." So far the system has a vocabulary of only 250 words. But there are no foreseeable limits to the number of words that the device can store or the number of languages it can be directed to translate.

Scholars and scientists who worked on it believe that within a few years the system may greatly increase communication, particularly in technical subjects, by making translation quick, accurate and easy.

The demonstration was at the headquarters of the International Business Machines Corporation, 590 Madison Avenue. It is the result of cooperative research by scientists of the corporation and scholars of the Georgetown University Institute of Languages and Linguistics in Washington.

The "mechanical" part of the translation system, which is mostly electronic, is a standard commercial model of the largest International Business Machines "stock" computer. This device, called the IBM Type 701 Electronic Data Processing Machine, was put on the market last April. Since then twelve of the machines have been sold to commercial, military and university computation laboratories.

The "literary" part of the system is a mechanical model of language devised at Georgetown by Prof. Leon Dostert and Dr. Paul Garvin. The corporation's share in the project was conducted by Dr. Cuthbert C. Hurd, director of its Division of Applied Science.

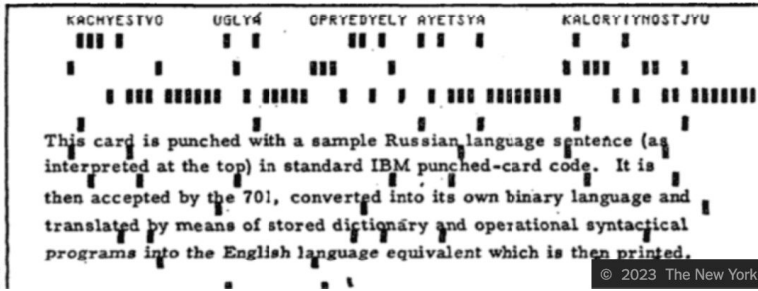
In the demonstration, a gir

Continued on Page 5, Column 2



The New York Tim

An electronic calculator produced by the International Business Machines Corporation wa demonstrated yesterday in a new role: translating Russian phrases into English. The de vice has a "vocabulary" of 250 Russian words and can be adapted to other languages. Mis Marilyn Polle "types" Russian phrases on I. B. M. punch cards that are fed into the machine



This card is punched with a sample Russian language sentence (as interpreted at the top) in standard IBM punched-card code. It is then accepted by the 701, converted into its own binary language and translated by means of stored dictionary and operational syntactical programs into the English language equivalent which is then printed.

Jan 8, 1954

© 2023 The New York

Machine Translation Era 1954-1966

- After the 1954 demo MT projects won massive grants for about a decade following
- Went into translating Russian into English
- The CIA was funding projects that amounted to 1.3 million dollars through the National Science Foundation



Russian to English

- Understanding and translating Russian language became a driving force for Machine Translation projects.
- A matter of national security in the throes of the Cold War.
- Russian was also increasingly the global language of science.
- By 1958, Russian abstracts made up (17%) of all entries in *Chemical Abstracts* trailing behind English (50%), the most comprehensive source of information on foreign abstracts.
- MIT translated German, a lab in the U.K. translated French, but the vast majority of Anglophone researchers worked on Russian.

“We chose Russian because we believe that today it is of very great importance to be able to communicate with the Russians … to make faster progress toward the goal of world peace.” – Thomas Watson, Jr. (chairman of IBM)

Russian Science

- Scientists were most interested in *scientific* Russian. esp natural sciences
- In 1956, Georgetown received substantial National Science Foundation fundings to undertake intensive research for the translation of Russian scientific materials into English
- These projects focused on translation scientific journals.
- By 1960, five separate gov agencies (NSF, CIA, the Army, Navy, and Air Force) were funding machine translation.

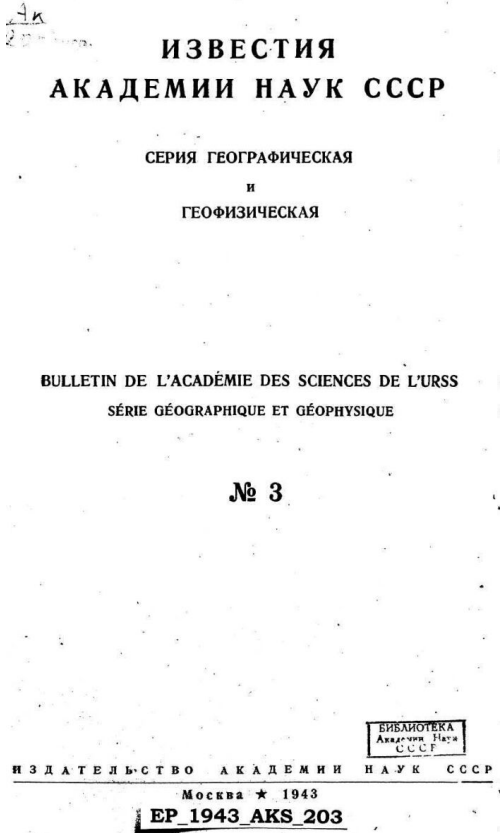




NSF backed translation project 1956

NSF backed the largest scientific translation project in the world of this time (1956)




ИЗВЕСТИЯ
АКАДЕМИИ НАУК СССР

СЕРИЯ ГЕОГРАФИЧЕСКАЯ
и
ГЕОФИЗИЧЕСКАЯ

BULLETIN DE L'ACADÉMIE DES SCIENCES DE L'URSS
SÉRIE GÉOGRAPHIQUE ET GÉOPHYSIQUE

№ 3

БИБЛИОТЕКА Академии Наук СССР

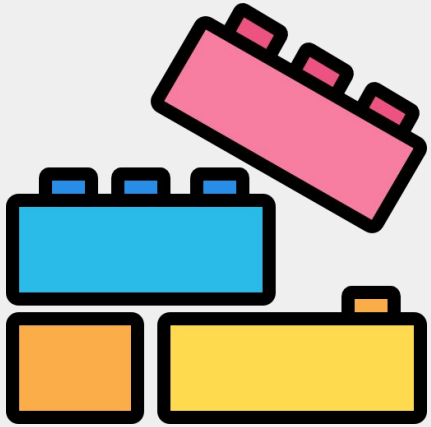
ИЗДАТЕЛЬСТВО АКАДЕМИИ НАУК СССР
Москва ★ 1943

EP_1943_AKS_203

Rule-based Shortcomings!

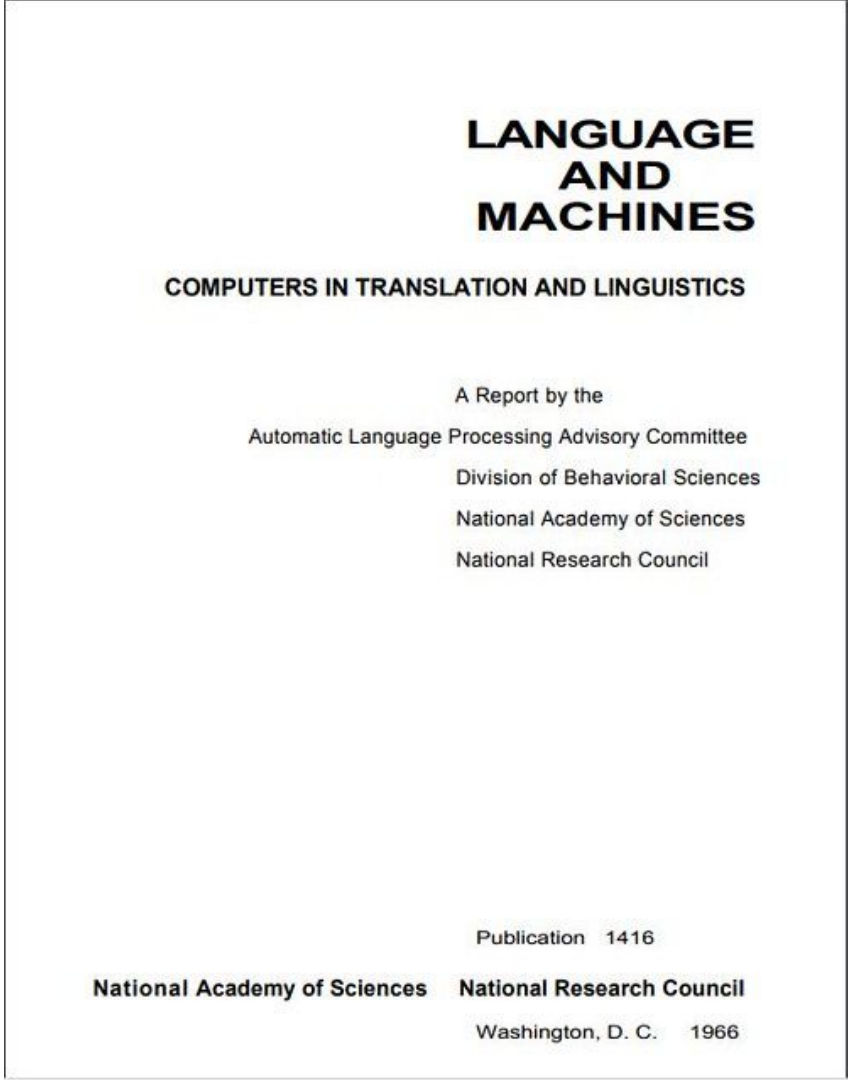
- Rule-based “toy problem” microworlds did not scale
- Language translation hit a ceiling
- Dictionary based machine translation was insufficient for fluent translation. Programs needed encyclopedia knowledge to understand natural language but there was neither software nor hardware to carry this capacity in the 1960s → **In 1966, all U.S. gov support for machine translation ended.**

ENG: “the spirit is willing but the flesh is weak”
RUS: “the vodka is strong but the meat is rotten”



Triggers First AI Winter (1970s)

"There is no immediate or predictable prospect of useful machine translation" (Automatic Language Processing Advisory Committee 1966)



LANGUAGE AND MACHINES

COMPUTERS IN TRANSLATION AND LINGUISTICS

A Report by the

Automatic Language Processing Advisory Committee

Division of Behavioral Sciences

National Academy of Sciences

National Research Council

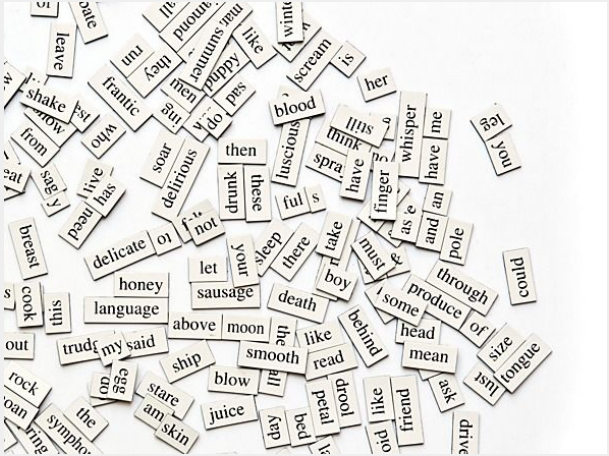
Publication 1416

National Academy of Sciences National Research Council

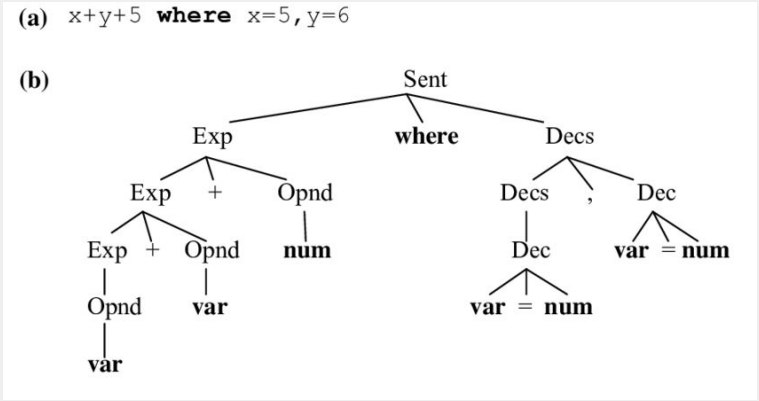
Washington, D. C. 1966

Empirical vs. Rational

- Contest in Language AI between probabilistic modeling vs. structuralism, or “data” empiricism vs. theoretical “rationalism”



Observation



Intuition & Deduction

Probabilistic Language Modeling 1940s & 1950s

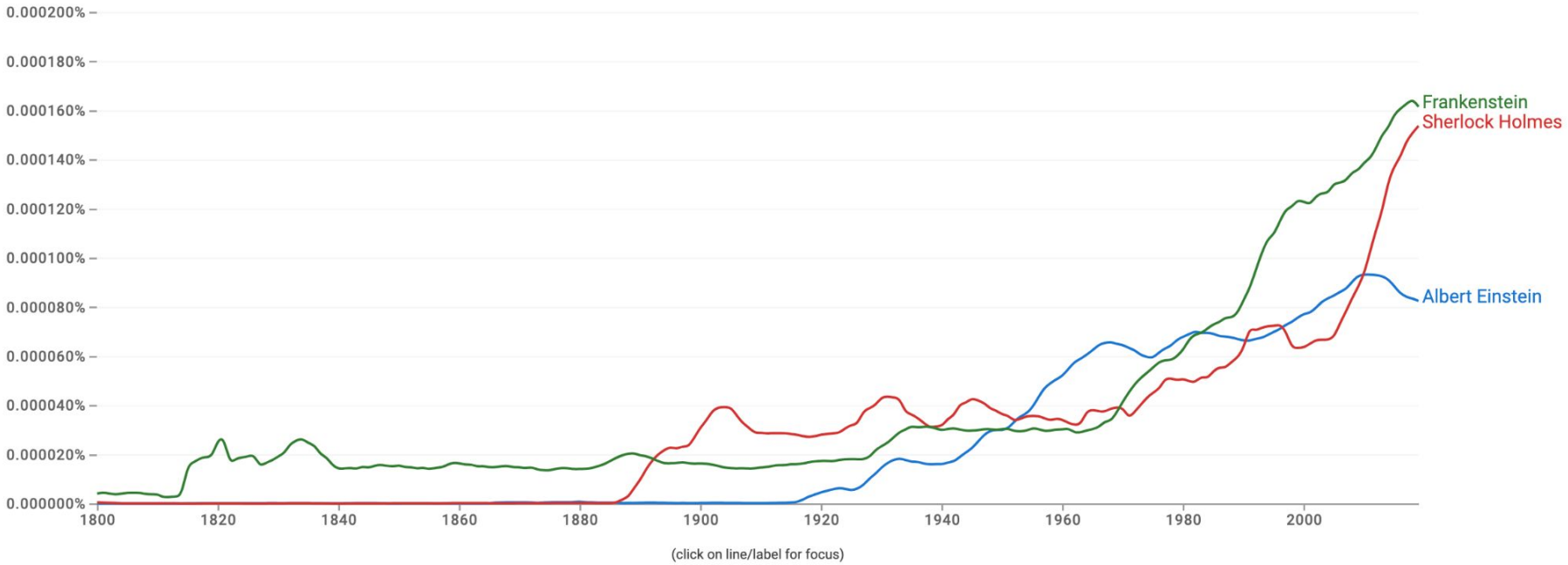
- Markov proposed n-gram letter models in 1913
- In 1949, Claude Shannon introduced the concept of n-gram language modeling for English

of times “My dog ate my” >= 5
of times “My dog ate my homework” = 5

$$P(\text{“homework”} \mid \text{“My dog ate my”}) = \frac{\text{Count(“My dog ate my homework”)}}{\text{Count(“My dog ate my”)}}$$

$$P(\text{“mother”} \mid \text{“My dog ate my”}) = \frac{\text{Count(“My dog ate my mother”)}}{\text{Count(“My dog ate my”)}}$$

Google N-grams



N-gram Frequencies

- Linguistics of the 1940s and 1950s focused on word and n-gram frequencies.

In 1957, linguist John Firth proclaimed

"You shall know a word by the company it keeps"

1-grams: "you", "shall", "know", "a", "word" ...
2-grams: "you shall", "shall know", "know a", "a word" ...
3-grams: "you shall know", "shall know a", "know a word" ...
4-grams: "you shall know a", "shall know a word", "know a word by"

Hardware Constraints



IBM 701

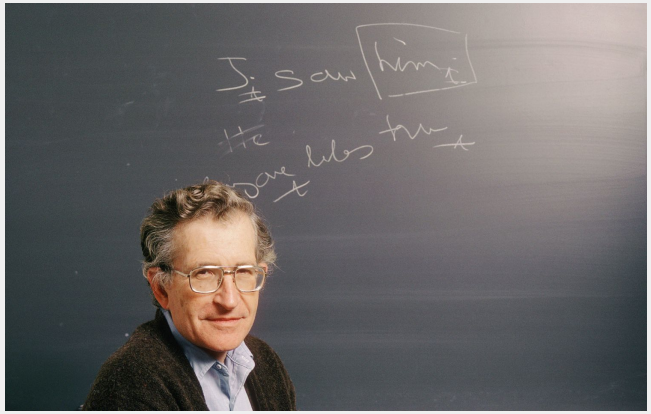


IBM PC

Years	Typical RAM	Examples
1950s	1-4 KB	IBM 701, UNIVAC
1960s	4-64 KB	IBM 360
1970s	64-512 KB	semiconductor chips
1980s	~640 KB	APPLE II, IBM PC
1990s	4-16 MB	
2000s	~4 GB	Wide adoption of PCs

Noam Chomsky's Context Free Models 1957/58

- Noam Chomsky introduced the idea of context-free models (and the Chomsky Hierarchy) and showed language has a logical structure!
- Started a movement to study syntax and disregard probabilistic, statistical approaches to language modeling
- For twenty years (1950s to 1970s), this approach dominated computational linguistics until empirical methods made a come back with statistical speech recognition



Palindrome of {a,b} Context Free Grammar

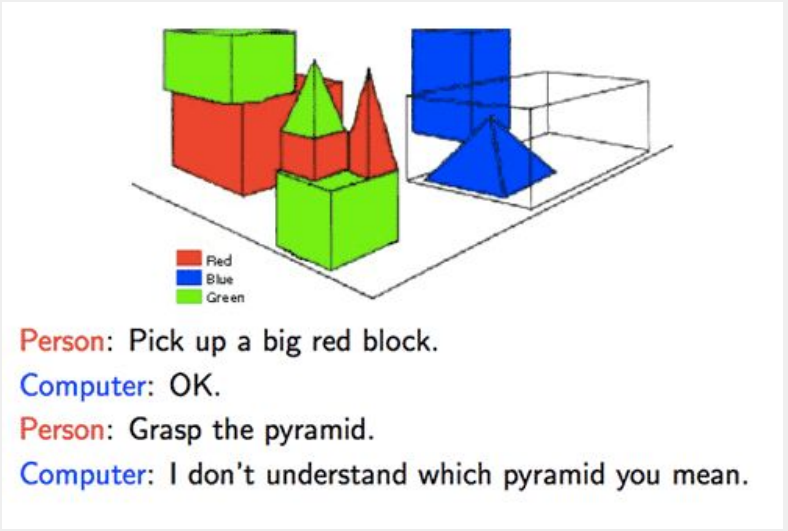
$$S \longrightarrow aSa \mid bSb \mid a \mid b \mid \langle end \rangle$$

start variable

terminal/nonterminal

Logic and rule-based methods 1960s →

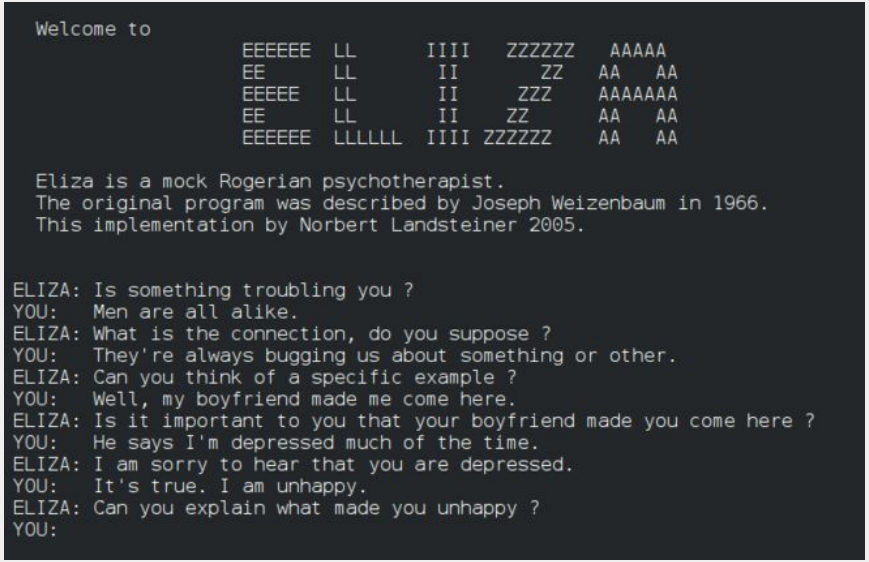
- Can’t capture all variables in the real world
- Best simulation is a microworld!



Terry Winograd’s microworld SHRDLU (1972)

ELIZA (1964-1967) by Joseph Weizenbaum

- First "chatterbot" to attempt Turing Test
- Directives in LISP-based pattern matching language MAD-SLIP that runs "scripts"
- DOCTOR script: Rogerian psychologist mimic
- Parody of "the responses of a non-directional psychotherapist in an initial psychiatric interview" and to "demonstrate the communication between man and machine was superficial"



ELIZA!

PARRY 1972 by Kenneth Colby

- More advanced than ELIZA
- Simulating a person with paranoid schizophrenia
- Passes the Turing Test in 1970s? (48%) – psychiatrists given transcripts of patient–doctor conversations
- International Conference on Computer Communication (ICCC) in 1973: ELIZA and PARRY interacted online on the ARPAnet!

PARRY -- In my opinion, you have a one-track mind.

ELIZA -- What makes you think I have a one-track mind?

PARRY -- I used to go to the track every week.

ELIZA -- What does that suggest to you?

PARRY -- I avoid racetracks now because the races are fixed.

ELIZA -- Is that the real reason?

PARRY -- I'd rather not discuss it.

ELIZA -- What else would you like to discuss?

PARRY -- That's something I don't want to go into."

Transcript of PARRY & ELIZA (DOCTOR)

Danny Bobrow's GUS (1977)

- GUS– Genial Understander System
- Intended to engage a sympathetic and highly cooperative human in an English dialog
- Restricted to the role of a travel agent in a conversation with a client who wants to make a *simple return trip to a single city in California*

GUS: Hello. My name is GUS. I can help you plan a simple trip by air. Where do you want to go?
(1) *Client:* I want to go to San Diego on May 28
GUS: What time do you want to leave?
(2) *Client:* I must be in San Diego before 10 am
GUS: Would you like P.S.A. flight 102 that arrives at 9.15 am?
(3) *Client:* What is the next flight?
GUS: Air California flight 310 that leaves at 8.30 am and arrives at 10.00 am
(4) *Client:* I'll take the first one
GUS: In what name should I make a reservation, please?
(5) *Client:* Dan Bobrow
GUS: I have confirmed the following flight:
P.S.A. flight 102 on Wednesday May 28 from San Jose to San Diego that leaves at 7.30 am and arrives at 9.15 am.
What date do you want to return on?
(6) *Client:* On Friday in the evening.
GUS: Would you like the flight that leaves at 7.45 pm?
(7) *Client:* That's fine.
GUS: I have confirmed the following flight:
P.S.A. flight 307 on Friday May 30 from San Diego to San Jose that leaves at 7.45 pm and arrives at 9.30 pm
Thank you for calling. Goodbye.

FIG. 1. A transcript of an actual dialog with GUS.

Adding a database! Question & Answering

- Question Answering systems were one of the first “task” solving NLP systems.
- The first Q&A system was BASEBALL proposed by Green et al in 1961 - a program made to handle questions about baseball (by accessing a database of baseball statistics).
- In 1973, another Q&A system followed called LUNAR by Woods et al – a program that answered questions about rocks brought back from the moon by the Apollo program.

BASEBALL 1961

In general, a question may contain implicit or explicit queries. Since these queries must be answered one at a time, several searches, with intermediate processing, are required. The first search operates on the stored data while successive searches operate on the found list generated by the preceding search operation. As an example, consider the question "On how many days in July did eight teams play?" The spec list is

```
Day (number of)  = ? ;
Month            = July;
Team (number of) = 8 .
```

On the first pass, the implicit question which teams is answered. The spec list for the first search is

```
Day   = Each;
Month = July;
Team  = ? .
```

The found data is a list of days in July; for each day there is a list of teams that played on that date. Following this search, the processor counts the teams for each day and associates the count with the attribute Team. On the second search, the spec list is

```
Day              = ?   ;
Month            = July;
Team (number of) = 8 .
```

The found data is a list of days in July on which eight teams played. After this pass, the processor counts the days, addsthe count to the found list and is finished.

Progress in natural language understanding—An application to lunar geology

by W. A. WOODS

Bolt Beranek and Newman Inc.
Cambridge, Mass.

INTRODUCTION

The advent of computer networks such as the ARPA net (see e.g., Ornstein et al.[3]) has significantly increased the opportunity for access by a single researcher to a variety of different computer facilities and data bases, thus raising expectations of a day when it will be a common occurrence rather than an exception that a scientist will casually undertake to use a computer facility located 3000 miles away and whose languages, formats, and conventions are unknown to him. In this foreseeable future, learning and remembering the number of different languages and conventions that such a scientist would have to know will require significant effort—much greater than that now required to learn the conventions of his local computing center (where other users and knowledgeable assistance is readily available). The Lunar Sciences Natural Language Information System (which we will hereafter refer to as LUNAR) is a research prototype of a system to deal with this and other man-machine communication problems by adapting the machine to the conventions of ordinary natural English rather than requiring the man to adapt to the machine.

State of the art

Although the state of the art in "understanding" natural language by machine is still very limited, significant advances in this area have been made in recent years. Since Simmons' first survey of question answering systems,[4] our understanding of the mysterious "semantic interpretation" component has been made more clear by work such as Woods,[7,8] and the techniques for mechanically parsing natural language sentences have been advanced by the advent of transition network grammars and their parsing algorithms.[9,10,11] Recently, Terry Winograd's blocks world system[6] has demonstrated the potential of some of these techniques—especially those of procedural semantics. The field is now at the point where prototype applications to real problems can make significant contributions to our understanding of the problems of natural language communication with machines. It must be realized, that such applications are still essentially research vehicles, since the problems of mechanical understanding of natural language remain far from solution. However, by using real problems, rather than imaginary toy problems, as the vehicles for such research, one cannot only focus the effort on problems in need of solution, but may also reap the additional benefit of producing a system which will perform a task

(GIVE ME THE AVERAGE SM ANALYSIS OF TYPE A ROCKS)
(WHAT IS THE AVERAGE MODAL CONCENTRATION OF ILMENITE IN TYPE A ROCKS?)
(GIVE ME EU DETERMINATIONS IN SAMPLES WHICH CONTAIN ILM)
(GIVE ME ALL K / RB RATIOS FOR BRECCIAS).
(WHAT BE ANALYSES ARE THERE?)
(GIVE ME OXYGEN ANALYSES IN S10084)
(WHAT SAMPLES CONTAIN CHROMITE?)
(WHAT SAMPLES CONTAIN P205?)
(GIVE ME THE MODAL ANALYSES OF P205 IN THOSE SAMPLES)
(GIVE ME THE MODAL ANALYSES OF THOSE SAMPLES FOR ALL PHASES)

```
31**(HOW MANY BRECCIAS CONTAIN OLIVINE)
***
PARSING
815 CONSES
4.633 SECONDS
INTERPRETING
1514 CONSES
7.29 SECONDS
INTERPRETATIONS:
(FOR THE X12 / (SEQL (NUMBER X12 / (SEQ TYPECS) : (CONTAIN X12 (NPR* X14 / (QUOTE OLIV)) (QU
NIL)))) : T ; (PRINOUT X12))

BBN LISP-10 03-09-72 . . .
EXECUTING
(5)

-----
***
32**(WHAT ARE THEY)
***
PARSING
487 CONSES
2.755 SECONDS
INTERPRETING
1158 CONSES
4.053 SECONDS
INTERPRETATIONS:
(FOR EVERY X12 / (SEQ TYPECS) : (CONTAIN X12 (NPR* X14 / (QUOTE OLIV)) (QUOTE NIL)) ; (PRINT
X12))

BBN LISP-10 03-09-72 . . .
EXECUTING
S10019
S10059
S10065
S10067
S10073
-----
***
34**(DO ANY SAMPLES HAVE GREATER THAN 13 PERCENT ALUMINUM)
***
PARSING
981 CONSES
4.614 SECONDS
INTERPRETING
902 CONSES
3.566 SECONDS
INTERPRETATIONS:
(TEST (FOR SOME X16 / (SEQ SAMPLES) : T ; (CONTAIN' X16 (NPR* X17 / (QUOTE AL203)) (GREATERT
13 PCT))))

BBN LISP-10 03-09-72
EXECUTING
YES.
T
```

LUNAR 1973

Expert Systems 1960s → 1980s

- From generality (General Problem Solver) to specific domain
- Capture a well-defined human expert knowledge domain (e.g. medicine, chemical spectrography)
- Encode "knowledge" as facts or rules!
- Restricting domains helps minimize common sense knowledge constraint for encoding



DENDRAL late 1960s

- DENDRAL was an expert system designed to discover molecular structure from mass spectra
- Collaborated with Djerassi (expert in mass spectrography) for more expert domain knowledge
- “Mother of all Expert Systems”?
- Moved to the SUMEX facility for national use (by all chemists!)

Elsevier

ARTINT 983

DENDRAL and Meta-DENDRAL: roots of knowledge systems and expert system applications

Edward A. Feigenbaum

Knowledge Systems Laboratory, Department of Computer Science, Stanford University, Stanford, CA 94305, USA

Bruce G. Buchanan

Computer Science Department, University of Pittsburgh, Pittsburgh, PA 15260, USA

During AI's first decade (1956–1966), the task environments in which AI scientists investigated their basic science issues were generally idealized “clean” task environments, such as propositional calculus theorem proving and puzzle solving. After the mid-1960s, a bolder and more applied inclination to choose complex real-world problems as task environments became evident. These efforts were both successful and exciting, in two ways. First, the AI programs were achieving high levels of competence at solving certain problems that human specialists found challenging (the excitement was that our AI techniques were indeed powerful and that we were taking the first steps toward the dream of the very smart machine). Second, these complex real-world task environments were proving to be excellent at stimulating basic science questions for the AI science, in knowledge representation, problem solving, and machine learning. To recognize and illuminate this trend, the *Artificial Intelligence Journal* in 1978 sponsored a special issue

MYCIN early 1970s

- A program that could identify bacteria that could cause infections and make treatment (antibiotic) recommendations
- It was accurate in about 69% of cases which was better than human disease experts when based on the same criteria (457 rules took 20 person years)

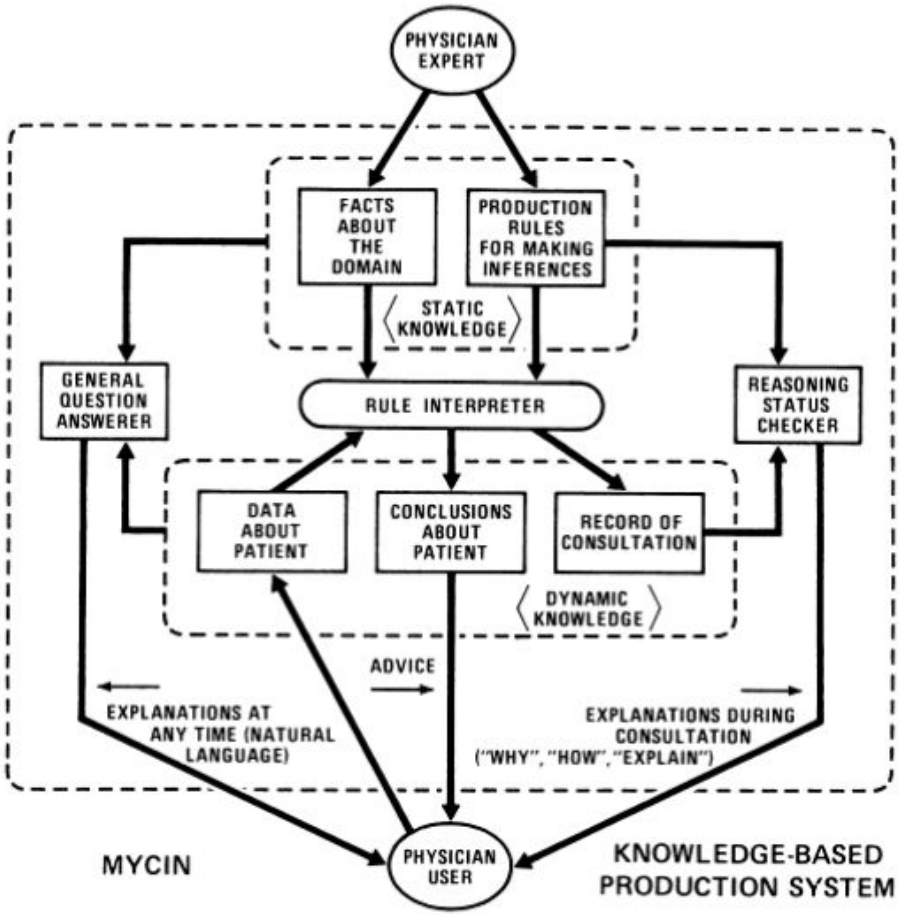


Figure 2 - Diagram summarizing the organization and flow of information within MYCIN. The correlation between this design and the human consultation process depicted in Fig. 1 is discussed in the text. (Figure reproduced from reference 10).

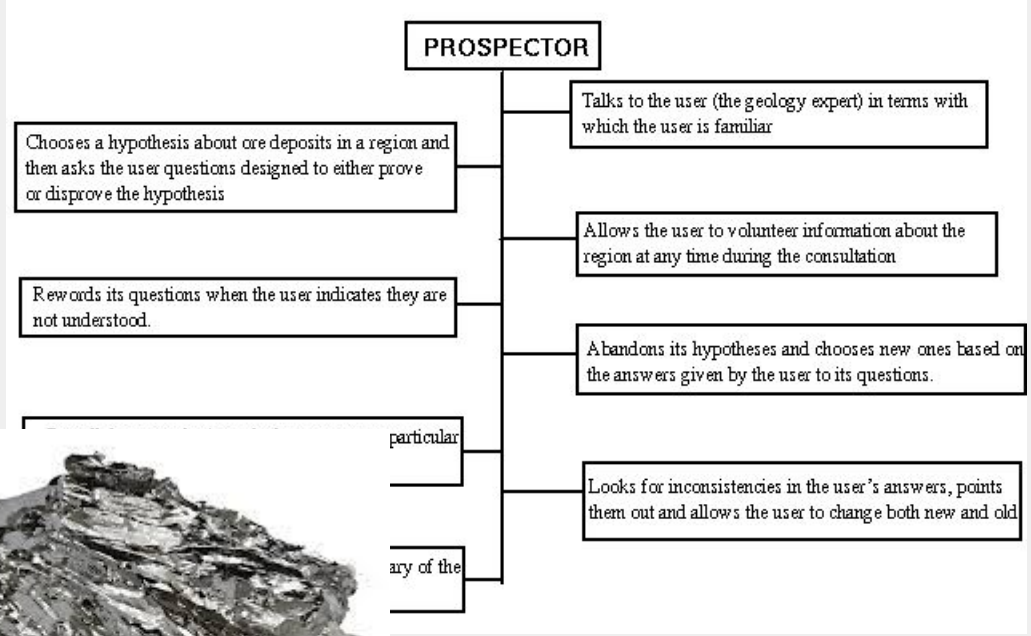
XCON or R1 (eXpert CONfigurer) 1980

- XCON was an system developed by DEC (digital equipment corporation) designed to help the computer manufacturer configure its machine to suit customer demand optimally
- First large-scale industrial use of AI
- 2500 rules
- $25m cost savings a year by reducing errors in order config and assembly process
- By 1986, had processed 80,000 orders and achieved an accuracy rate of 95-98%
- Good example of AI-led productivity → inspired 1980s investment



PROSPECTER 1983

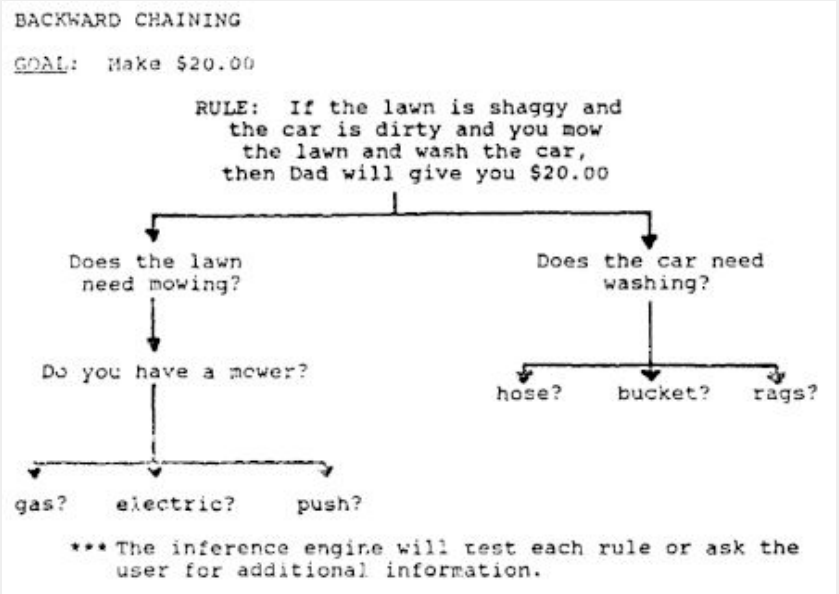
- PROSPECTOR was a geology expert system that helped discover new drilling sites
- Made with 9 geologists
- Helped a company discover molybdenum deposits in Washington!





Expert Systems have common structure

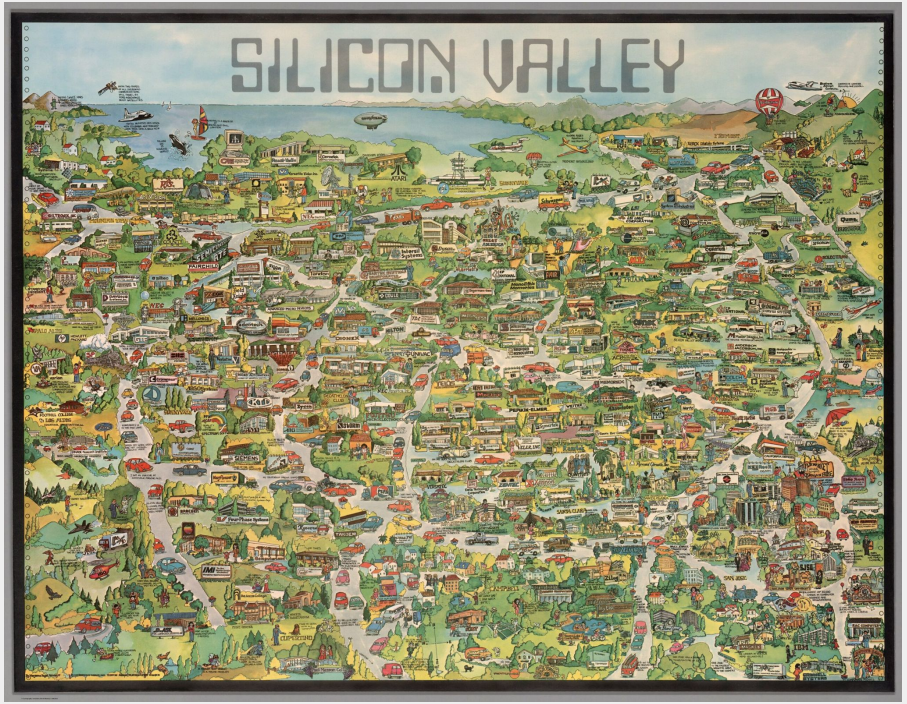
1. the knowledge base
   (i.e. list of IF____THEN____)
2. short-term memory that contains specific data about the actual problem
3. tree-searching
   (i.e. inference machine)
4. user interface
   (e.g. menu, quasi-natural language dialogue)



How to make this system more generic??

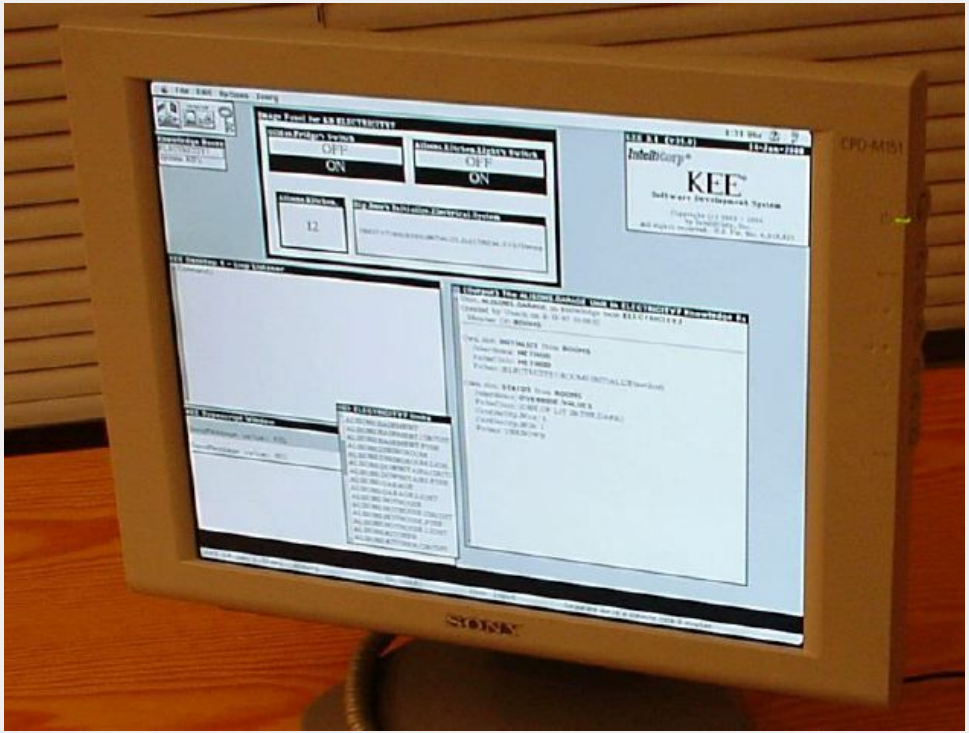
Expert Systems Start-ups 1980s

- Expert Systems was the first AI technology to arouse commercial interest
- Teknowledge, Inc., Intellicorp sprang up in the early 1980s to market these systems. Both these companies received millions from DARPA to bring AI into the market and military fields.

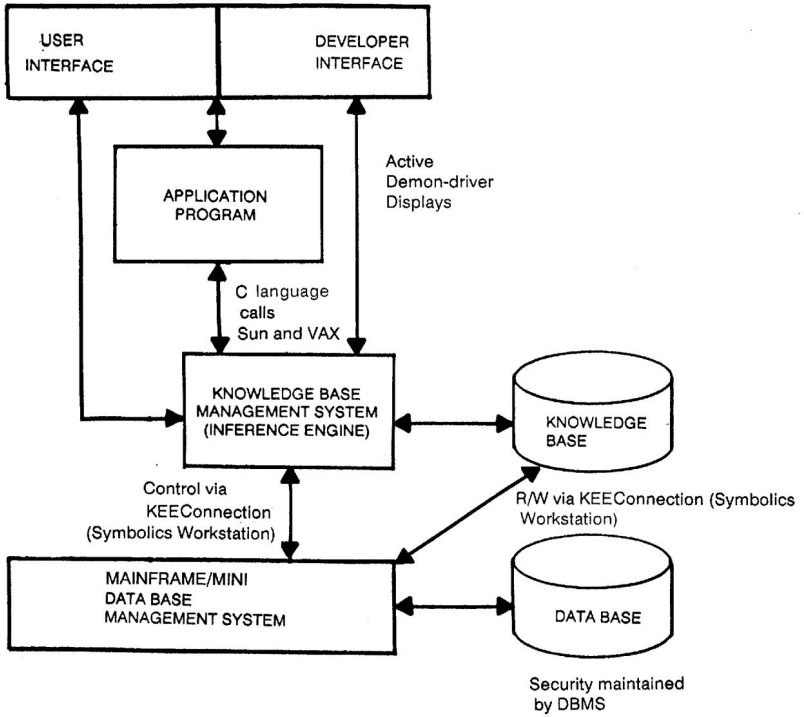


Knowledge Engineering Environment (KEE) 1983

- Intellicorp created and started selling Knowledge Engineering Environment (KEE) 1983: a high-end knowledge expert system development shell.
- KEE was to support knowledge acquisition and representation, reasoning, and user interface construction.
- Framework for multiple expert domains



Product Information



Software

- **KEE Version 3.0 for workstations**
- **KEE Version 3.0 for 80386-based IBM-compatible PCs**
- **KEEConnection**
- **SimKit**
- **Run-Time KEE**

KEE Versions 1.0 & 2.0 and Intelliscope are no longer sold.

Pricing

KEE Version 3.0 (workstations)	**$46,500** academic price **$9,250**
KEE Version 3.0 (386 PCs)	**$9,900**
KEEConnection (varies with hardware)	**$15,500** academic price **$5,000**
SimKit	**$21,000** academic price **$4,125**
Run-Time KEE (single-user workstation)	**$5,000**

Pricing varies, depending on number of licenses purchased, and includes 3-9 days training for 2 persons and one year's support. Support services include standard consultants, senior scientist consultants and/or apprenticeship training (4 weeks).

Hardware

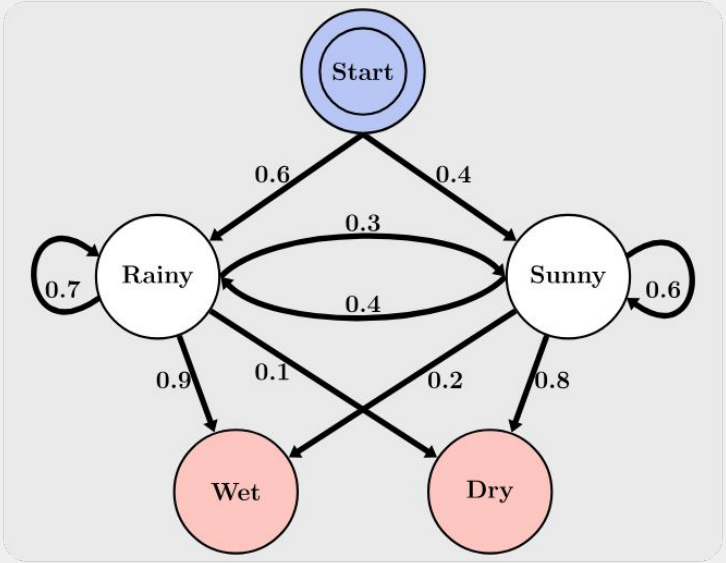
all 80386-based IBM PC-compatible computers
IBM PC RT
Symbolics 36XX series
Xerox 1100 series
Sun 3 workstations
Apollo Domain workstations
TI Explorer I and II
DEC VAX series
MicroVAX II workstations

DARPA disappointed in Language AI again

- In the late 1980s, DARPA began cutting back bankrolling Teknowledge and Intellicorp projects
- AI commercial products failed to meet expectations of investors
- Second AI winter ensues

Speech goes statistical

- 1971 DARPA funded speech recognition projects
- 4 separate and competing projects over 5 years
- CMU made the winning project: HAPPY (based on DRAGON) built on Hidden Markov Models
- Hidden Markov Models used commonly for integrating sounds with words sequence + probability
- DARPA disappointed



Strategic Computing Initiative 1983-93 & Speech

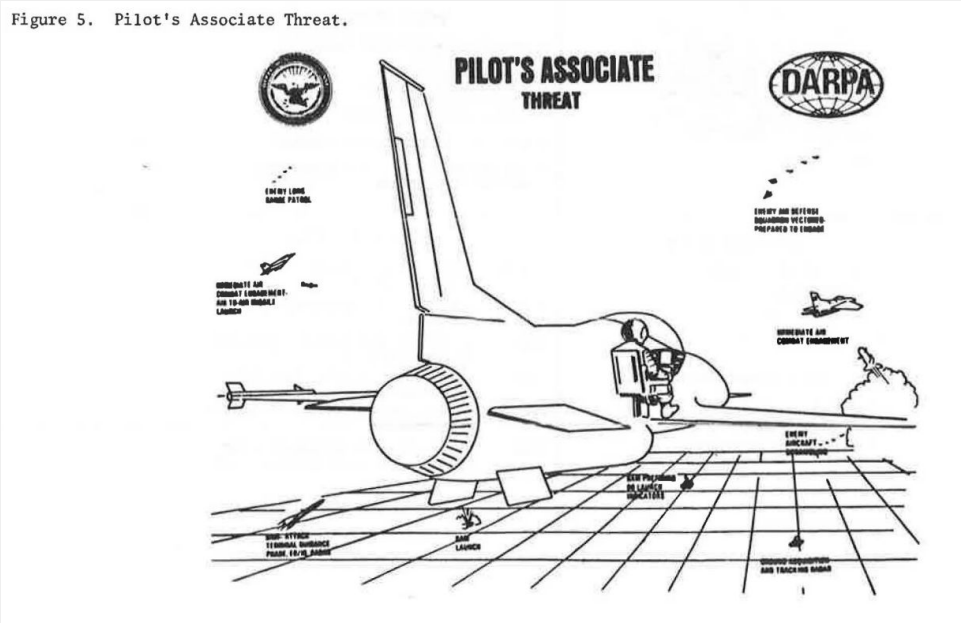
- Carnegie Mellon and MIT researchers in 1978 and 1979 concluded that phonetic signals were more tractable than previously believed.
- Started two SCI funded projects on speech recognition
- Project 1) Phone recognition: It turns out spoken English could be condensed to just 48 discrete sounds (and silence). These sounds could be converted to a stream of data a computer could interpret.
- Project 2) n-gram stats in speech: researchers began to integrate n-gram stats to model word order to disambiguate similar/same sounding words (“write” vs. “right”)

I have to “write”/”right” this paper
p(“write this paper”) vs. p(“right this paper”)

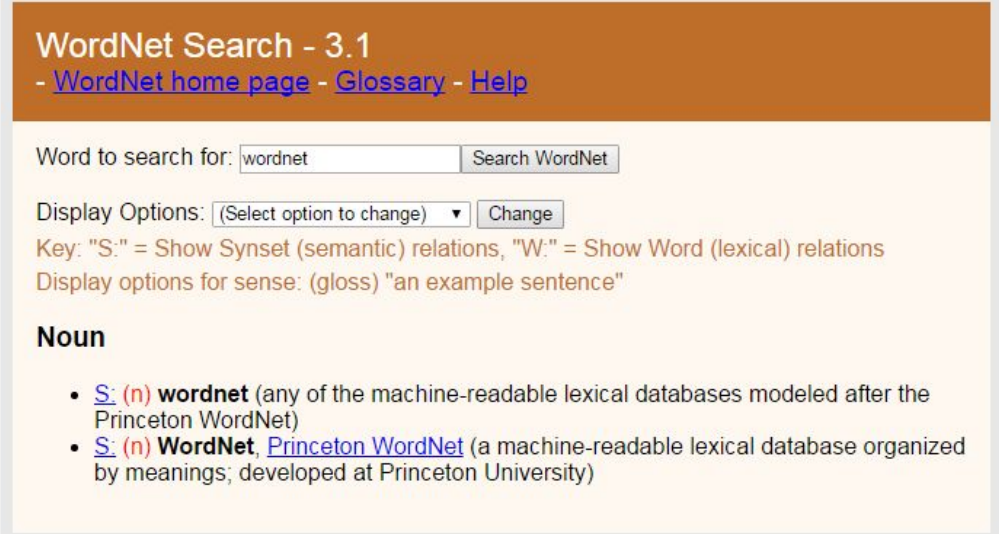
Speech in Defense

- DARPA wanted to integrate speech into their two military projects: Pilot's Associate and Battle Management.
- It funded 8 contractors in nine projects in continuous speech to Carnegie Mellon for Battle Management.
- And 5 contractors in 8 projects for robust speech to Texas Instruments for Pilot's Associate.
- By the end of the Strategic Computing Initiative, the continuous speech recognition program could recognize 10,000 words in a moderately noisy background → started being integrated into military and commercial applications.

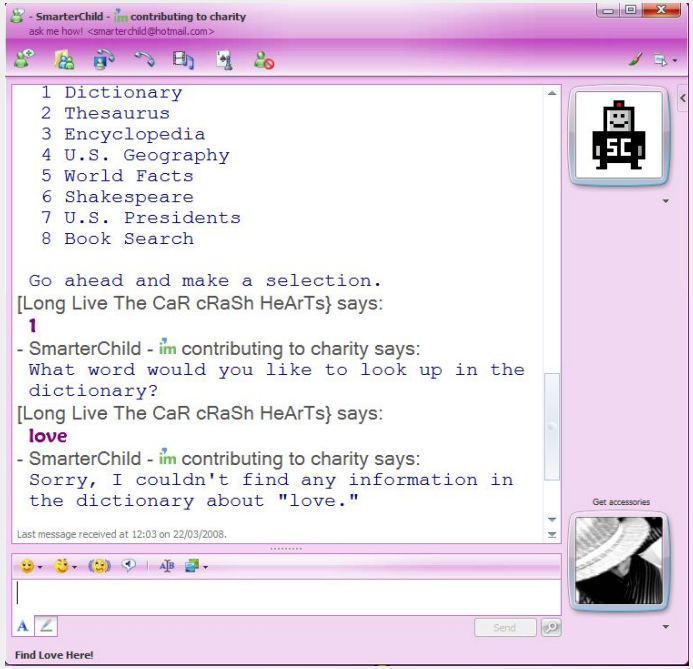
Figure 5. Pilot's Associate Threat.



Internet & Big Data 1992 →



Wordnet 2001



2001 SmarterChild on AOL Instant Messenger (AIM) or MSN Messenger

A publicly available dictionary of 100,000 words and phrases (labeled with parts of speech tags and linked by semantic relations such a synonyms, antonyms)



The Stanford Integrated Digital Library Project

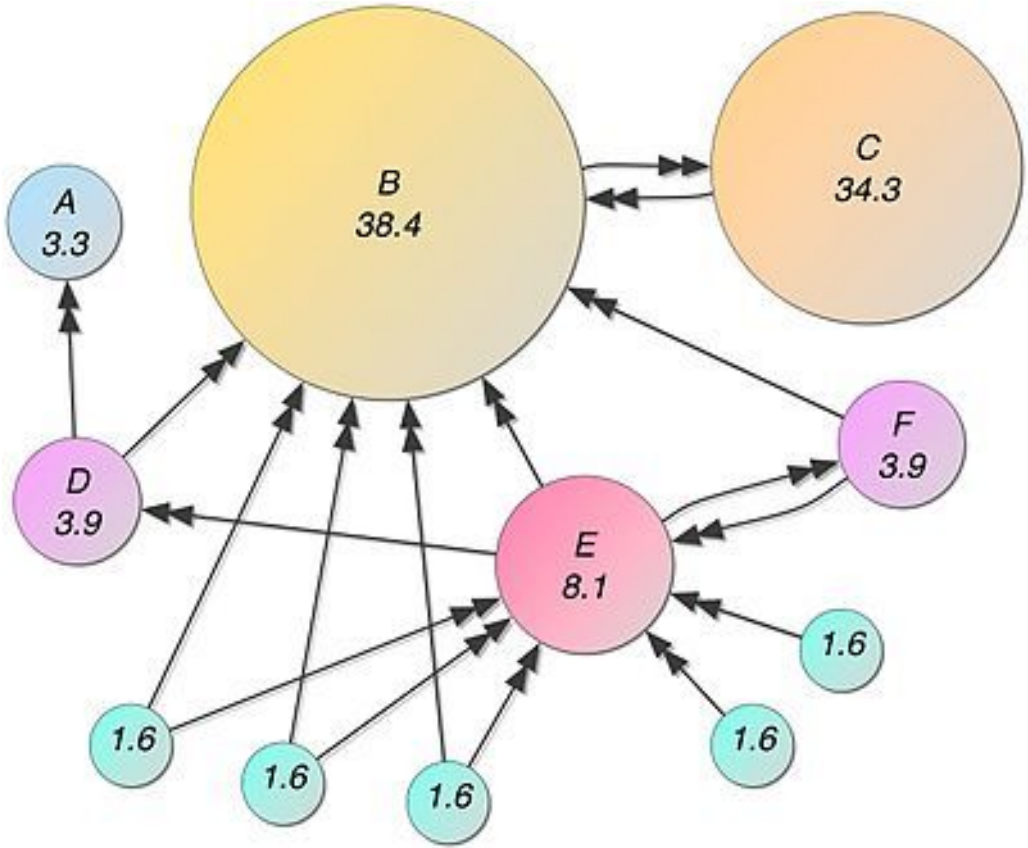
Summary

The goal of the Stanford Integrated Digital Library Project (SIDLP) is to develop the enabling technologies for a single, integrated and "universal" library, providing uniform access to the large number of emerging networked information sources and collections. These include both on-line versions of pre-existing works and new works and media of all kinds that will be available on the globally interlinked computer networks of the future. The Integrated Digital Library is broadly defined to include everything from personal information collections, to the collections that one finds today in conventional libraries, to the large data collections shared by scientists. Our technology will provide the "glue" that will make this worldwide collection usable as a unified entity, in a scalable and economically viable fashion.

A consortium of information vendors, publishers, end users, commercializers, and researchers has been formed to carry out the proposed research. They will also implement a SIDLP Testbed system that will provide access to significant collections of computing and related literature. The testbed will be used to develop a variety of novel library services, and to experiment with protocols for the integration of heterogeneous information sources.

NSF Award Abstract 9411306. October 5, 1998

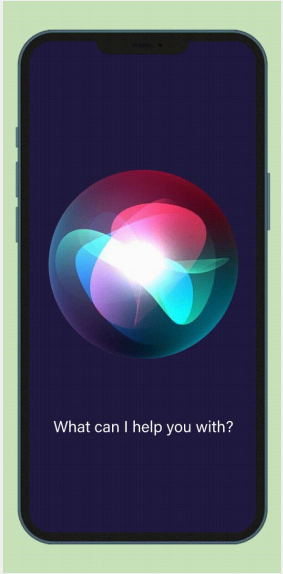
PageRank algorithm was introduced in 1998 (Brin and Page) as a way to search documents on the web.



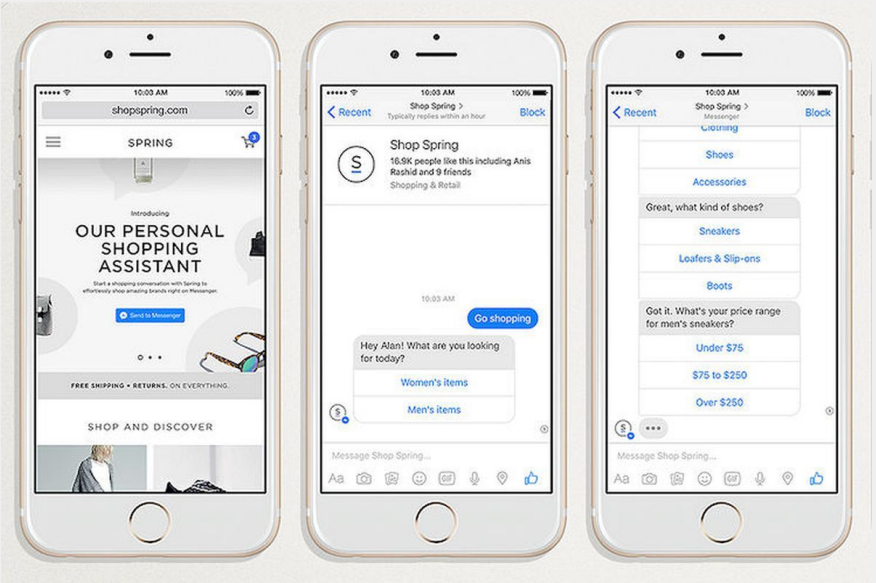
Commercialization



2000 Google AdWords



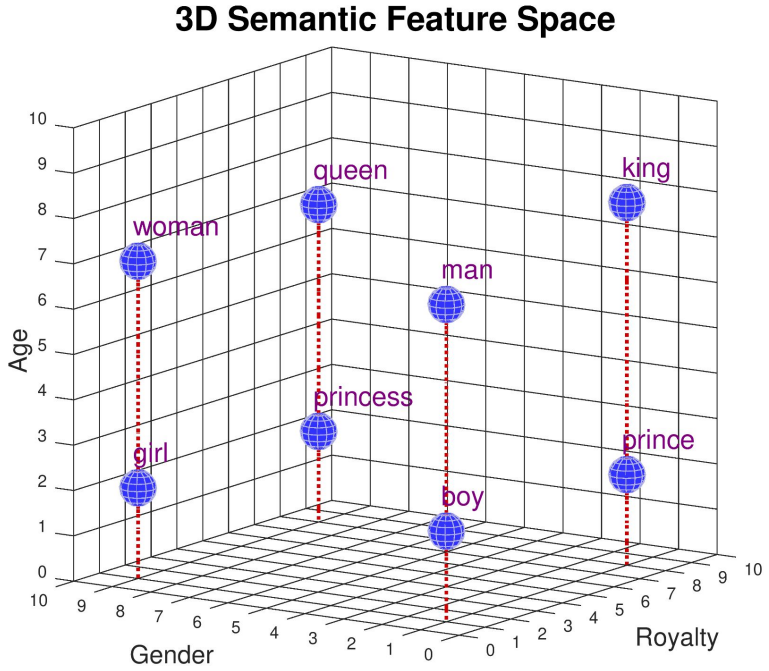
2010 Siri



2016 Bots for Messenger (Bot-plication)

More data → Empirical methods

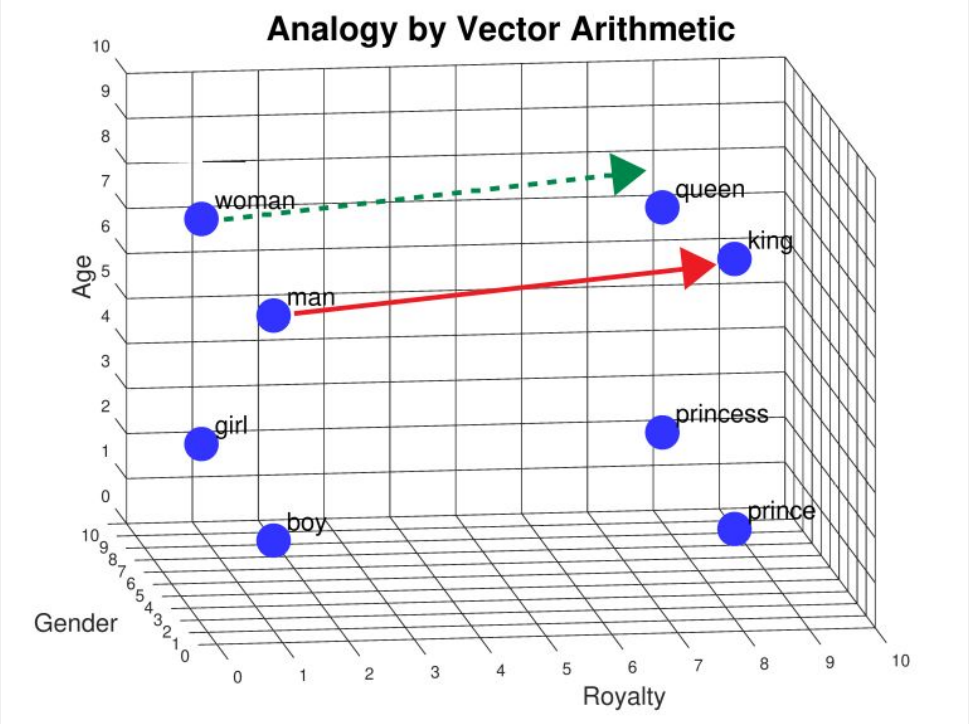
Word Coordinates	Gender	Age	Royalty
man	[1,	7,	1]
woman	[9,	7,	1]
boy	[1,	2,	1]
girl	[9,	2,	1]
king	[1,	8,	8]
queen	[9,	7,	8]
prince	[1,	2,	8]
princess	[9,	2,	8]



https://www.cs.cmu.edu/~dst/WordEmbeddingDemo/tutorial.html

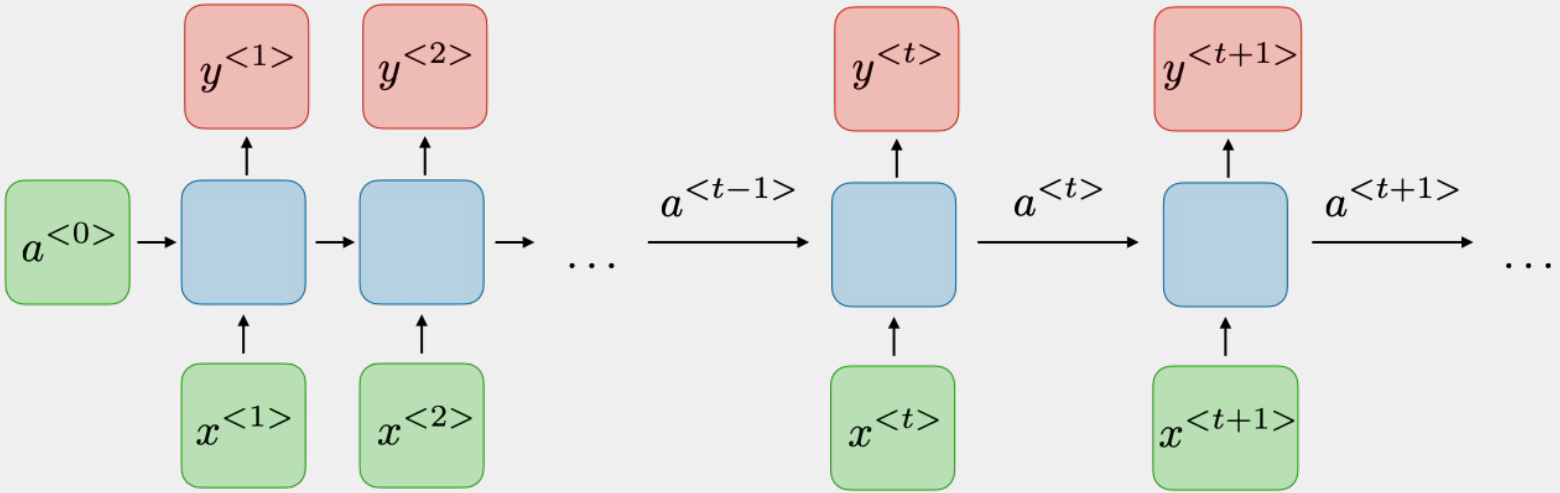
More data → Empirical methods

king	[	1,	8,	8]
man	[	1,	7,	0]
king - man	[	0,	1,	8]
woman	[	9,	7,	0]
king - man + woman	[	9,	8,	8]
queen	[	9,	7,	8]



https://www.cs.cmu.edu/~dst/WordEmbeddingDemo/tutorial.html

Sequential Modeling with Neural Nets



https://stanford.edu/~shervine/teaching/cs-230/cheatsheet-recurrent-neural-networks

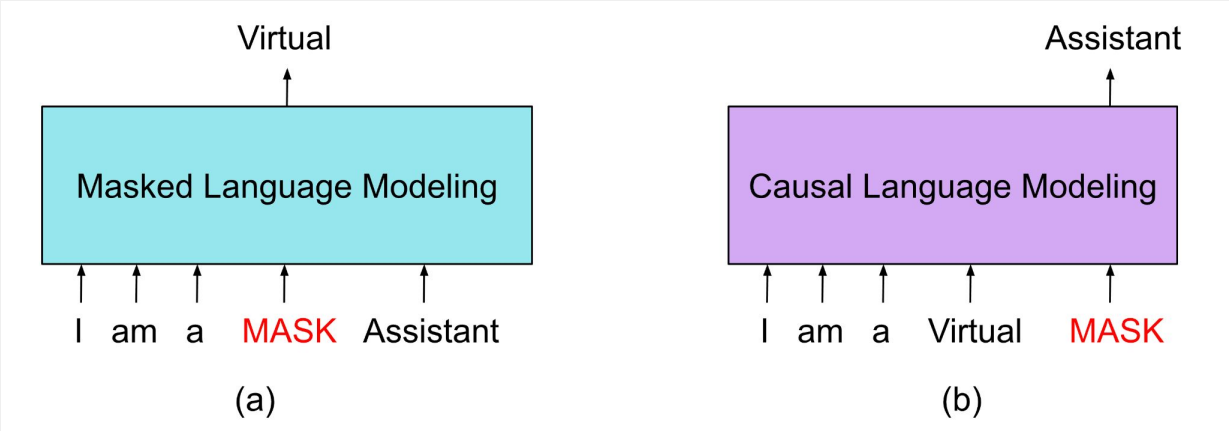
Language Modeling with Unlabeled Text

1. Masked Language Modeling

Training the model to correctly predict missing (masked) words in a given sequence (encoder models)

2. Autoregressive (causal) Language Modeling

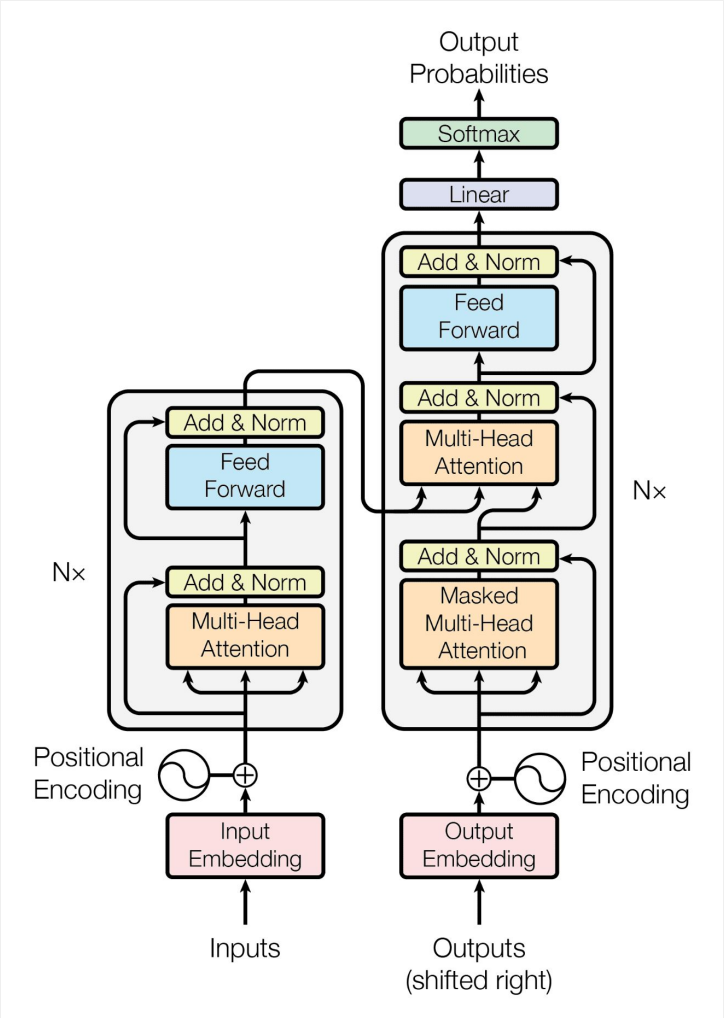
Training the model to learn by optimizing predicting the next word (GPT models)



Transformer Model 2016/2017

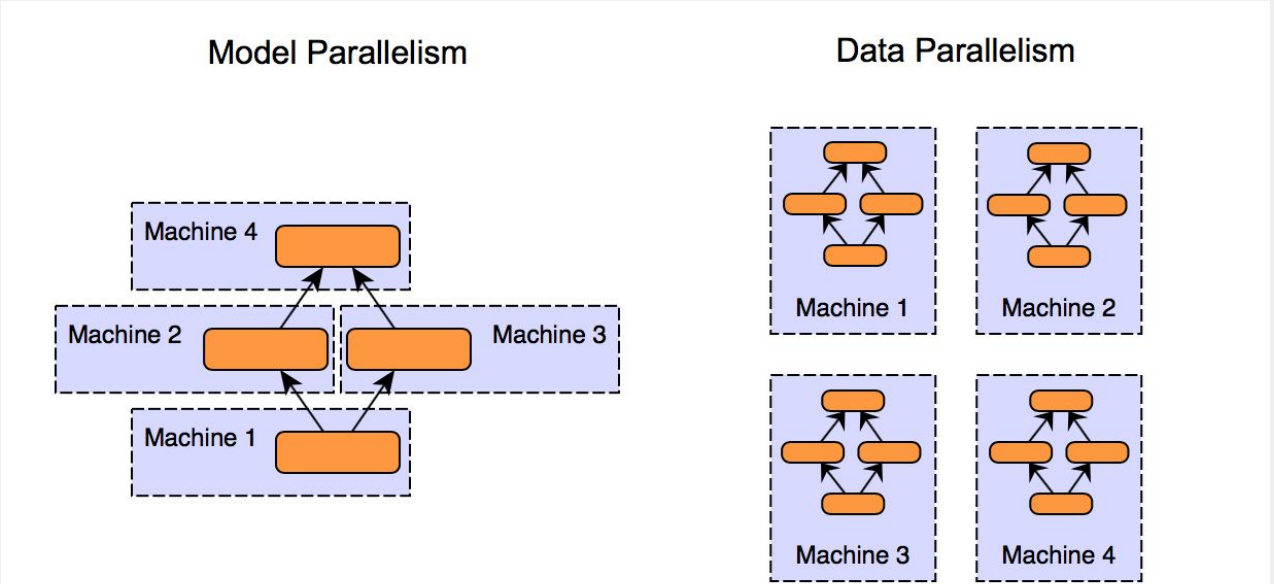
- Highly effective for encoding and decoding sequential/structured information

1. Attention mechanism for long-range dependencies
2. Parallel training across multiple machines



Parallel Training

https://pytorch.org/torchrec/high-level-arch.htmll

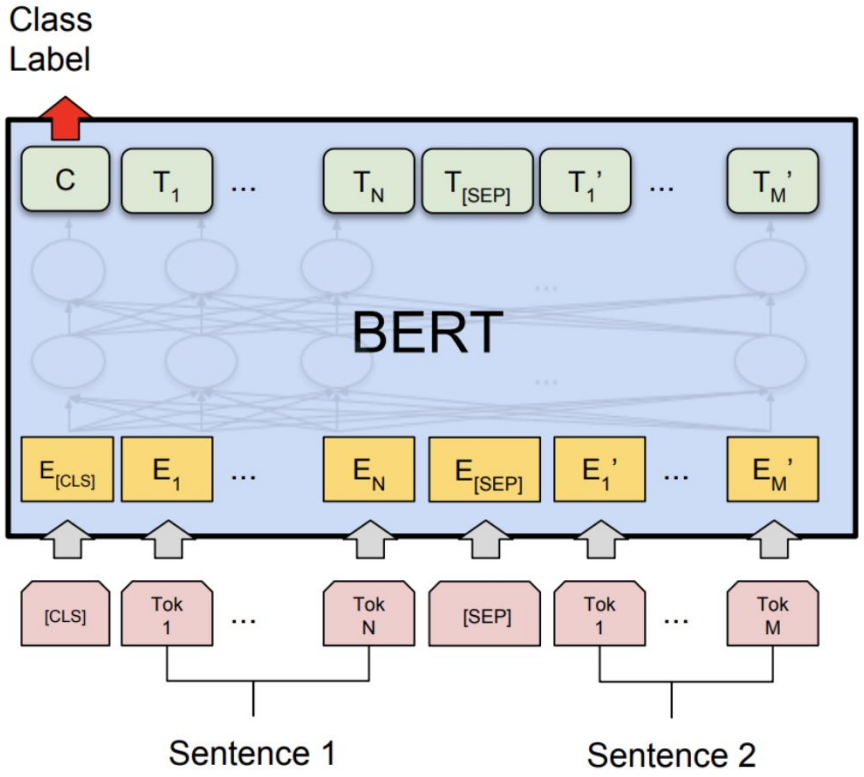


- Split model weights (Embedding Dimension)
- Distribute model across GPUs
- Good for models with lots of weights
- Need to change code
- Only Transformers

- Split data (Batch Size)
- Each GPU computes → Shares loss
- RNNs & Transformers
- Good for training with lots of data

Bidirectional Encoder Representations 2018





Data vs. Model Size

Though there has been significant recent work allowing larger and larger models to be trained, our analysis suggests an increased focus on dataset scaling is needed. Speculatively, we expect that scaling to larger and larger datasets is only beneficial when the data is high-quality. This calls for responsibly collecting larger datasets with a high focus on dataset quality. Larger datasets will require extra care to ensure train-test set overlap is properly accounted for, both in the language modelling loss but also with downstream tasks. Finally, training for trillions of tokens introduces many ethical and privacy concerns. Large datasets scraped from the web will contain toxic language, biases, and private information. With even larger datasets being used, the quantity (if not the frequency) of such information increases, which makes dataset introspection all the more important. *Chinchilla* does suffer from bias and toxicity but interestingly it seems less affected than *Gopher*. Better understanding how performance of large language models and toxicity interact is an important future research

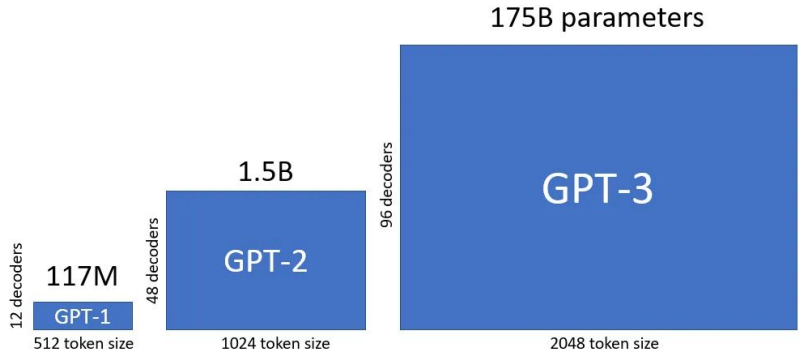
ology towards the training of auto-regressive language models, de-off between model size and the amount of data in other very expensive, choosing the optimal model size and training thods we propose are easy to reproduce in new settings.

Table 1 | **Current LLMs**. We show five of the current largest dense transformer models, their size, and the number of training tokens. Other than LaMDA (Thoppilan et al., 2022), most models are trained for approximately 300 billion tokens. We introduce *Chinchilla*, a substantially smaller model, trained for much longer than 300B tokens.

Model	Size (# Parameters)	Training Tokens
LaMDA (Thoppilan et al., 2022)	137 Billion	168 Billion
GPT-3 (Brown et al., 2020)	175 Billion	300 Billion
Jurassic (Lieber et al., 2021)	178 Billion	300 Billion
Gopher (Rae et al., 2021)	280 Billion	300 Billion
MT-NLG 530B (Smith et al., 2022)	530 Billion	270 Billion
Chinchilla	70 Billion	1.4 Trillion

Training Compute-Optimal Large Language Models. March 2022. DeepMind

OpenAI's Transformer Models



GPT-1
- Created in 2018
- Have a data set of 117 million parameters
- Performs tasks without labeled data

GPT-2
- Created in 2019
- Has a data set of 1.5 billion parameters
- Trained on a more diverse set of data than GPT-1

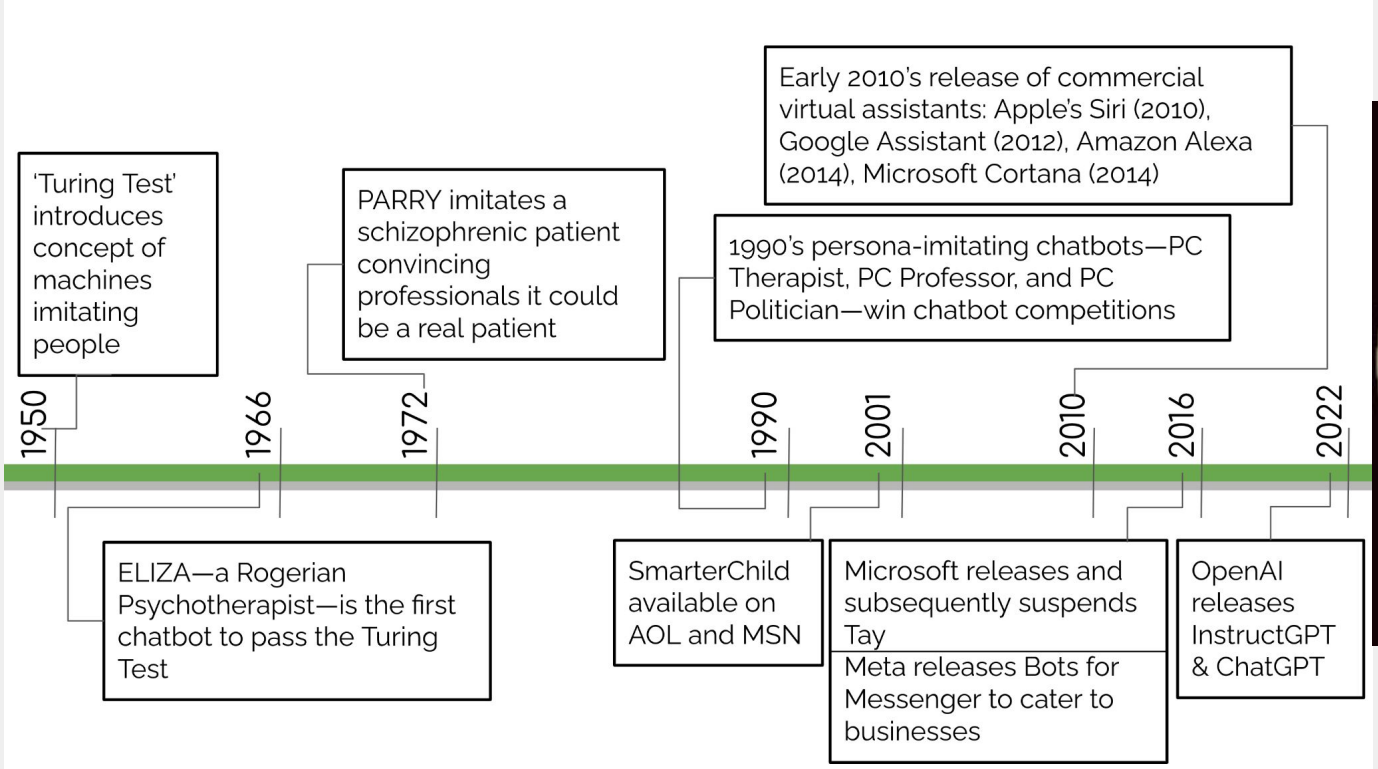
GPT-3
- Created in 2020
- Have a data set of 175 million parameters
- Solves simple arithmetic problems

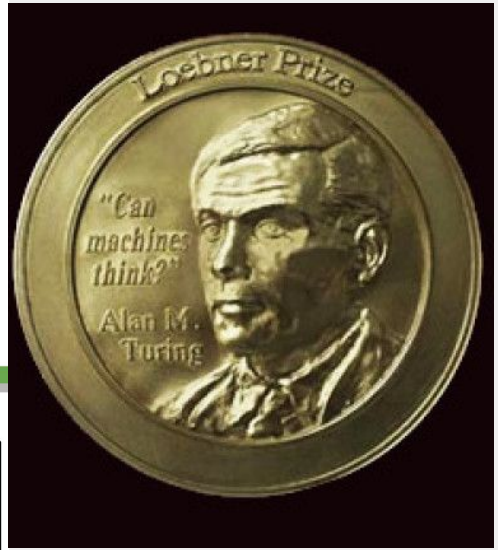
ChatGPT(GPT-3.5)
- Created in 2022
- Can create letters, song lyrics, research papers, recipes, therapy sessions, poems, essays, outines and software code

GPT-4
- Created in 2023
- Multimmodal AI which means you can interact with it in speech as well as textual interactions

OpenAI

Chatbots: From Turing Test to ChatGPT





Voice: How far we've come!

https://elevenlabs.io/app/speech-synthesis/text-to-speech

